

From Product Hilbert Spaces to the Generalized Koopman Operator and the Nonlinear Fundamental Lemma

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Abstract—The generalization of the Koopman operator to systems with control input and the derivation of a nonlinear fundamental lemma are two open problems that play a key role in the development of data-driven control methods for nonlinear systems. In this paper we derive a novel solution to these problems based on basis functions expansion in a product Hilbert space constructed as the tensor product between the Hilbert spaces of the state and input observable functions, respectively. We identify relaxed invariance conditions that guarantee existence of a bounded linear operator, i.e., the generalized Koopman operator, from the constructed product Hilbert space to the Hilbert space corresponding to the lifted state propagated forward in time. Compared to classical Koopman invariance conditions, measure preservation is not required. Moreover, we derive a nonlinear fundamental lemma by exploiting the constructed exact infinite-dimensional bilinear Koopman representation and Hankel operators. The effectiveness of the developed generalized Koopman embedding is illustrated on the Van der Pol oscillator and in predictive control of a soft-robotic manipulator model.

Index Terms—Data-driven modeling and control, Hilbert space, Koopman operator, Nonlinear dynamical systems, Willems' fundamental lemma.

I. INTRODUCTION

THE development of data-driven analysis and control methods for dynamical systems is currently fueled by the merging of machine learning and control design methods, see, e.g., [1], [2], which can potentially impact all control application domains in our society. The main challenge in developing data-driven analysis and control methods lies in dealing with the inherently complex and nonlinear dynamics of real-world systems. Two of the most adopted approaches in data-driven analysis and control of nonlinear systems are the Koopman operator modeling approach [1], [3] and the behavioural modeling approach based on Willems' fundamental lemma [2], [4]. In what follows, we provide a non-exhaustive overview of recent advances and open problems in these two areas and we show that they share a common challenge, which will be addressed in this paper.

The Koopman operator modeling approach [1], [3], [5], [6] constructs an infinite-dimensional linear representation of a nonlinear system that can be used for analysis or control.

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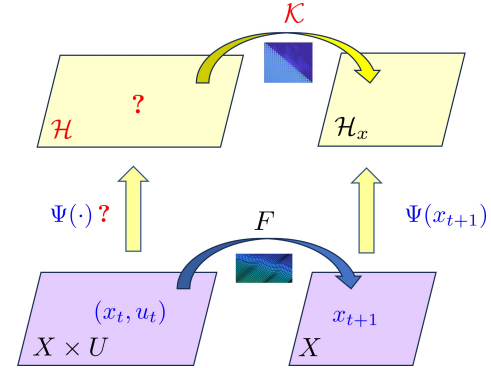


Fig. 1: Illustration of the Koopman operator modeling approach for dynamical systems with control input.

In the case of autonomous systems, that is, $x_{t+1} = F_0(x_t)$ with $F_0 : X \rightarrow X$ a nonlinear map, this is typically done by means of a measure-preserving composition operator $(C_{F_0}\varphi)(x) = \varphi(F_0(x))$, i.e., the Koopman operator, which acts on observable functions φ in a Hilbert space $\mathcal{H}_x$. In practice, finite-dimensional approximations of the Koopman operator can be computed from data, see, e.g., the extended dynamic mode decomposition (EDMD) method [7]. Recently, it was shown in [8] that Koopman operators can be constructed using a direct encoding method based on orthonormal expansion in Hilbert spaces. However, when the Koopman operator approach is applied to dynamical systems with control input (that is $x_{t+1} = F(x_t, u_t)$), as graphically illustrated in Figure 1, a new problem emerges: how to construct the Hilbert space $\mathcal{H}$ of observable functions on which the operator acts?

One of the first approaches to deriving a Koopman operator for systems with inputs originated in [9], which augments the system state with an infinite-dimensional input sequence and constructs a measure-preserving composition operator for the augmented autonomous system. Therein it was shown how this approach can be used to derive approximate linear and bilinear Koopman representations. A similar approach was also employed in [10]. Conditions for exact Koopman bilinearization (in the original control input) of input-affine nonlinear systems were derived in [11], [12] and further researched, e.g., in [13], [14]. Other works derive a family of (autonomous) Koopman operators parameterized by the control input [15], [16], or derive linear-parameter-varying Koopman representations [17].

As an alternative to the Koopman modeling approach, Willems' fundamental lemma [2], [4] aims at deriving a data-based system representation defined as a basis of recorded trajectories whose span coincides with all possible system trajectories under the assumption of persistency of excitation (PE). However, this result exploits the superposition principle, which is only valid for linear systems. Under a stronger PE condition, fundamental lemmas have been derived for state-space Hammerstein systems [18], bilinear systems in [19], [20] and input-output generalized bilinear systems in [21]. In [22], [23] it was shown that the fundamental lemma can be equivalently formulated using kernel functions and reproducing kernel Hilbert spaces (RKHS). Similarly as in the Koopman modeling approach, the key challenge lies in the construction of lifting basis functions such that in the lifted (Hilbert) space there exists a possibly infinite-dimensional, linear data-based system representation.

Motivated by the above state-of-the-art in Koopman-based and data-driven behavioral modeling of nonlinear systems, in this paper we develop a novel theoretical framework that uses *product Hilbert spaces* for generalizing the Koopman operator and Willems' fundamental lemma to nonlinear systems with inputs. The *main contributions* are as follows:

- A well-defined generalized Koopman composition operator for systems with inputs that acts on observable functions in a Hilbert space $\mathcal{H}_x$ defined on the state-space X and yields functions in the tensor product $\mathcal{H} := \mathcal{H}_x \otimes \mathcal{H}_u$ of Hilbert spaces defined on the state-space X and input-space U , respectively;
- A coordinate representation of the generalized Koopman operator in Riesz bases as an operator $\mathcal{K} : \ell^2 \rightarrow \ell^2$ that yields an exact infinite-dimensional bilinear representation of the original nonlinear system;
- A nonlinear fundamental lemma based on the infinite-dimensional bilinear Koopman representation (extended to systems with measured outputs) and Hankel operators.

The remainder of the paper is organized as follows. Preliminaries are introduced in Section II. The generalized Koopman operator based on product Hilbert spaces is presented in Section III. The nonlinear fundamental lemma is derived in Section IV. Illustrative examples are shown in Section V and conclusions are summarized in Section VI.

II. PRELIMINARIES

Let $\mathbb{C}$, $\mathbb{R}$ and $\mathbb{N}$ denote the sets of complex, real and natural numbers, respectively. Let $\mathbb{K}$ denote either $\mathbb{C}$ or $\mathbb{R}$. For $q \in \mathbb{N} \cup \{\infty\}$ vectors (or vector-valued functions) $\{v_1, \dots, v_q\}$, define $\text{col}(v_1, \dots, v_q) := [v_1^\top, \dots, v_q^\top]^\top$. For two matrices $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{p \times q}$, $A \otimes B \in \mathbb{R}^{mp \times nq}$ denotes their Kronecker product. For two vectors $a \in \mathbb{R}^n$, $b \in \mathbb{R}^m$, $a \otimes b \in \mathbb{R}^{nm}$ denotes their Kronecker vector product, i.e., for $a = [a_1 \ a_2]^\top$, $b = [b_1 \ b_2]^\top$, $a \otimes b = [a_1 b_1 \ a_1 b_2 \ a_2 b_1 \ a_2 b_2]^\top$. For two matrices $A \in \mathbb{R}^{n \times m}$, $B \in \mathbb{R}^{p \times m}$, $A \odot B \in \mathbb{R}^{np \times m}$ denotes their Khatri-Rao product, which consists of the column-wise Kronecker product. For a full-row rank matrix $A \in \mathbb{R}^{m \times n}$, $A^\dagger = A^\top (AA^\top)^{-1}$ denotes its right Moore-Penrose inverse. For any sequence $\mathbf{s} := \{s_0, s_1, \dots\}$ with $s_i \in \mathbb{R}^{n_s}$, define

the truncated augmented vector $\mathbf{s}_{[i,j]} := \text{col}(s_i, \dots, s_j)$ for $0 \leq i < j$. For a set $X \subseteq \mathbb{R}^n$ and $N \in \mathbb{N}_{\geq 2}$, X^N denotes the N -times Cartesian product $X \times \dots \times X$. The *Kronecker delta* δ_{ik} for indices $i, k \in \mathbb{N}$ is defined as

$$\delta_{ik} := \begin{cases} 1 & \text{if } i = k, \\ 0 & \text{if } i \neq k. \end{cases}$$

Let (X, Σ_x, μ_x) and (U, Σ_u, μ_u) be σ -finite measure spaces with $X \subseteq \mathbb{R}^n$ and $U \subseteq \mathbb{R}^m$, and let $(X \times U, \Sigma_x \otimes \Sigma_u, \mu_x \otimes \mu_u)$ denote their product measure space [24].

Definition II.1 Let $F : X \times U \rightarrow X$ be a measurable map. The *pushforward* of $\mu_x \otimes \mu_u$ under F yields the measure $F_*(\mu_x \otimes \mu_u)$ on (X, Σ_x) defined by

$$[F_*(\mu_x \otimes \mu_u)](A) := (\mu_x \otimes \mu_u)(F^{-1}(A)), \quad \forall A \in \Sigma_x, \quad (1)$$

where $F^{-1}(A) := \{(x, u) \in X \times U : F(x, u) \in A\} \in \Sigma_x \otimes \Sigma_u$ is the preimage of A under F .

In this paper we will follow standard notation for function spaces $L^p(X, \mu_x)$, see, e.g., [24, Chapter 3].

Assumption II.2 For every $A \in \Sigma_x$, let $\nu(A) := (\mu_x \otimes \mu_u)(F^{-1}(A))$ denote the pushforward measure through F . The pushforward measure ν through F is absolutely continuous with respect to the measure μ_x (denoted by $\nu \ll \mu_x$), i.e., it holds that $\mu_x(A) = 0 \implies \nu(A) = 0$.

The above assumption states that F does not map sets of positive $(\mu_x \otimes \mu_u)$ -measure in $X \times U$ into μ_x -null sets in X . Then, since (X, Σ_x, μ_x) is σ -finite and $\nu \ll \mu_x$, by the *Radon-Nikodym theorem* [24, Theorem 6.10] there exists a unique function $w \in L^1(X, \mu_x)$, $w \geq 0$ μ_x -a.e., called the *Radon-Nikodym derivative* $w := d\nu/d\mu_x$, such that

$$\nu(A) = \int_A w(x') d\mu_x(x'), \quad \forall A \in \Sigma_x. \quad (2)$$

Assumption II.3 The equality (2) holds with a function $w \in L^\infty(X, \mu_x)$, i.e., $M := \|w\|_{L^\infty(X, \mu_x)} < \infty$.

Assumption II.3 requires that μ_x -a.e., the preimage $F^{-1}(A)$ is bounded relative to $\mu_x(A)$ uniformly over measurable $A \in \Sigma_x$.

We will extensively use $L^2(X, \mu_x)$ and ℓ^2 spaces, for which we provide detailed definitions in what follows. Let

$$\ell^2 := \{(x_n)_{n \geq 1} : \sum_{n=1}^{\infty} |x_n|^2 < \infty\}, \quad \|(x_n)\|_{\ell^2} := \left(\sum_{n=1}^{\infty} |x_n|^2 \right)^{\frac{1}{2}}.$$

For $X \subseteq \mathbb{R}^n$ define $\mathcal{H}_x := L^2(X, \mu_x)$ as

$$\mathcal{H}_x = \{\varphi : X \rightarrow \mathbb{K} \text{ measurable} \mid \int_X |\varphi(x)|^2 d\mu_x(x) < \infty\},$$

where functions agreeing μ_x -almost everywhere are identified, i.e. $\varphi_1 = \varphi_2$ in $\mathcal{H}_x$ iff $\mu_x(\{x : \varphi_1(x) \neq \varphi_2(x)\}) = 0$.

Furthermore, define the inner product and norm as

$$\begin{aligned} \langle \varphi_1, \varphi_2 \rangle_{\mathcal{H}_x} &:= \int_X \varphi_1(x) \overline{\varphi_2(x)} d\mu_x(x), \\ \|\varphi\|_{\mathcal{H}_x} &:= \left(\int_X |\varphi(x)|^2 d\mu_x(x) \right)^{1/2}. \end{aligned}$$

Let a corresponding Hilbert space on $U \subseteq \mathbb{R}^m$, i.e., $\mathcal{H}_u := L^2(U, \mu_u)$, be defined similarly. Then we define the corresponding *product Hilbert space* as

$$\mathcal{H} := L^2(X \times U, \mu_x \otimes \mu_u) \quad (3)$$

with inner product and norm

$$\begin{aligned} \langle f_1, f_2 \rangle_{\mathcal{H}} &:= \int_{X \times U} f_1(x, u) \overline{f_2(x, u)} d(\mu_x \otimes \mu_u), \\ \|f\|_{\mathcal{H}} &:= \left(\int_{X \times U} |f(x, u)|^2 d(\mu_x \otimes \mu_u) \right)^{1/2}. \end{aligned} \quad (4)$$

Definition II.4 A sequence $\{\varphi_i\}_{i=1}^{\infty}$ in a Hilbert space $\mathcal{H}$ is an *orthonormal basis* (ONB) if $\langle \varphi_i, \varphi_k \rangle_{\mathcal{H}} = \delta_{ik}$ and $\overline{\text{span}}\{\varphi_i\} = \mathcal{H}$. It is a *Riesz basis* if there exist constants $0 < \alpha \leq \beta < \infty$ such that for all $(c_i)_{i \geq 1} \in \ell^2$:

$$\alpha \sum_{i=1}^{\infty} |c_i|^2 \leq \left\| \sum_{i=1}^{\infty} c_i \varphi_i \right\|_{\mathcal{H}}^2 \leq \beta \sum_{i=1}^{\infty} |c_i|^2. \quad (5)$$

Note that every ONB is a Riesz basis with $\alpha = \beta = 1$ and a Riesz basis is a bounded and boundedly invertible image of an ONB under a bounded linear operator [25, Theorem 3.6.6].

Furthermore, observe that the tensor product Hilbert space $\mathcal{H}_x \otimes \mathcal{H}_u$ is isometrically isomorphic to the product space $\mathcal{H}$:

$$L^2(X, \mu_x) \otimes L^2(U, \mu_u) \cong L^2(X \times U, \mu_x \otimes \mu_u), \quad (6)$$

via the identification $(\varphi \otimes \psi)(x, u) := \varphi(x)\psi(u)$, extended by linearity and continuity [26, Theorem II.10].

The following result follows directly from standard properties of product spaces and Riesz bases [26, Section II.4].

Proposition II.5 If $\{\psi_{i,x}\}_{i=1}^{\infty}$ is a Riesz basis for $\mathcal{H}_x$ with bounds α_x, β_x and $\{\psi_{i,u}\}_{i=1}^{\infty}$ is a Riesz basis for $\mathcal{H}_u$ with bounds α_u, β_u , then the product functions

$$\psi_{(i,l)}(x, u) := \psi_{i,x}(x) \psi_{l,u}(u), \quad i, l \in \mathbb{N}, \quad (7)$$

form a Riesz basis for $\mathcal{H}$ with bounds $\alpha_x \alpha_u \leq \beta_x \beta_u$.

III. THE GENERALIZED KOOPMAN OPERATOR

For discrete-time autonomous nonlinear systems,

$$x_{t+1} = F_0(x_t), \quad t \in \mathbb{N},$$

with the map $F_0 : X \rightarrow X$ Lebesgue integrable, the classical Koopman composition operator [3] is defined as

$$C_{F_0} : \mathcal{H}_x \rightarrow \mathcal{H}_x, \quad (C_{F_0}\varphi)(x) = \varphi(F_0(x)),$$

where $\varphi : X \rightarrow \mathbb{K}$ is called an observable (function). If F_0 is *measure preserving* and bijective, the invariance property

$$\varphi \circ F_0 \in \mathcal{H}_x, \quad \forall \varphi \in \mathcal{H}_x \quad (8)$$

holds, which guarantees that the Koopman composition operator is a well-defined bounded linear unitary operator.

In this paper we consider nonlinear dynamical systems with control input, i.e.,

$$x_{t+1} = F(x_t, u_t), \quad t \in \mathbb{N}, \quad (9)$$

where $F : X \times U \rightarrow X$ is a Lebesgue integrable map and $X \subseteq \mathbb{R}^n$, $U \subseteq \mathbb{R}^m$ denote the state space and the input space.

In what follows we adopt a product Hilbert space approach to define a generalized Koopman composition operator as

$$C_F : \mathcal{H}_x \rightarrow \mathcal{H}, \quad (C_F\varphi)(x, u) := \varphi(F(x, u)), \quad (10)$$

where $\mathcal{H} = \mathcal{H}_x \otimes \mathcal{H}_u$ is the product Hilbert space of the state and input Hilbert spaces, respectively. Note that now the generalized Koopman composition operator acts on observable functions $\varphi \in L^2(X, \mu_x)$ and yields functions $\varphi \circ F \in L^2(X \times U, \mu_x \otimes \mu_u)$. As such, this yields a novel, *generalized invariance* condition, i.e.,

$$\varphi \circ F \in \mathcal{H} = \mathcal{H}_x \otimes \mathcal{H}_u, \quad \forall \varphi \in \mathcal{H}_x. \quad (11)$$

Conditions on the map F that yield a well-defined generalized Koopman composition operator are given next.

Theorem III.1 Let (X, Σ_x, μ_x) and (U, Σ_u, μ_u) be σ -finite measure spaces and let $F : X \times U \rightarrow X$ be measurable. Define the pushforward $\nu := F_*(\mu_x \otimes \mu_u)$ on (X, Σ_x) as in Definition II.1 and suppose Assumptions II.2 and II.3 hold. Then the generalized Koopman composition operator

$$C_F : \mathcal{H}_x \rightarrow \mathcal{H}, \quad (C_F\varphi)(x, u) = \varphi(F(x, u)),$$

is a well-defined bounded linear operator with operator norm

$$\|C_F\|_{\mathcal{H}_x \rightarrow \mathcal{H}} := \sup_{\substack{\varphi \in \mathcal{H}_x \\ \|\varphi\|_{\mathcal{H}_x} = 1}} \|C_F\varphi\|_{\mathcal{H}} \leq M^{1/2}.$$

Proof: Linearity: For $\varphi_1, \varphi_2 \in \mathcal{H}_x$ and $\alpha, \beta \in \mathbb{K}$:

$$\begin{aligned} (C_F(\alpha\varphi_1 + \beta\varphi_2))(x, u) &= (\alpha\varphi_1 + \beta\varphi_2)(F(x, u)) \\ &= \alpha\varphi_1(F(x, u)) + \beta\varphi_2(F(x, u)), \\ &= \alpha(C_F\varphi_1)(x, u) + \beta(C_F\varphi_2)(x, u). \end{aligned}$$

Boundedness: Since F is measurable and φ is measurable, the composition $\varphi \circ F$ is measurable on $(X \times U, \Sigma_x \otimes \Sigma_u)$. For any $\varphi \in \mathcal{H}_x$, one can then apply the *change-of-variables formula* for pushforward measures [27, eq. (1.1)], i.e., if $T : (Y, \mathcal{B}) \rightarrow (Z, \mathcal{C})$ is measurable and λ is a measure on $\mathcal{B}$, then $\int_Z f d(T_*\lambda) = \int_Y f \circ T d\lambda$ for any non-negative measurable f . Here $T_*\lambda$ denotes the *pushforward* of the measure λ under the map T , defined by $(T_*\lambda)(A) := \lambda(T^{-1}(A))$ for measurable sets A (cf. Definition II.1). In our setting, $Y = X \times U$ with $\lambda = \mu_x \otimes \mu_u$, and $Z = X$ with $T_*\lambda = \nu$ the pushforward measure on X . Then the change-of-variables formula with $T = F$ and $f = |\varphi|^2 \geq 0$ yields

$$\begin{aligned} \|C_F\varphi\|_{\mathcal{H}}^2 &= \int_{X \times U} |(\varphi \circ F)(x, u)|^2 d(\mu_x \otimes \mu_u) \\ &= \int_X |\varphi(x')|^2 d\nu(x'). \end{aligned}$$

Substituting $d\nu = w d\mu_x$ from (2) further yields

$$\|C_F\varphi\|_{\mathcal{H}}^2 = \int_X |\varphi(x')|^2 w(x') d\mu_x(x').$$

Applying Assumption II.3, $w(x') \leq M$ for μ_x -a.e. x' gives

$$\|C_F\varphi\|_{\mathcal{H}}^2 \leq M \int_X |\varphi(x')|^2 d\mu_x(x') = M \|\varphi\|_{\mathcal{H}_x}^2 < \infty.$$

Taking the supremum over $\|\varphi\|_{\mathcal{H}_x} = 1$ yields $\|C_F\|_{\mathcal{H}_x \rightarrow \mathcal{H}} \leq M^{1/2}$. ■

Theorem III.1 establishes that under Assumptions II.2 and II.3, $C_F : \mathcal{H}_x \rightarrow \mathcal{H}$ is bounded with $\|C_F\| \leq M^{1/2}$, which implies $\varphi \circ F \in \mathcal{H}$ for every $\varphi \in \mathcal{H}_x$ since $\|\varphi \circ F\|_{\mathcal{H}}^2 = \|C_F \varphi\|_{\mathcal{H}}^2 \leq M \|\varphi\|_{\mathcal{H}_x}^2 < \infty$. This yields the generalized invariance condition (11). A few remarks are in order.

Remark III.2 (Relation with the Koopman operator) The classical [3] *Koopman invariance condition* (8) requires μ_x to be F_0 -invariant, i.e., $\mu_x(A) = \mu_x(F_0^{-1}(A))$, for all $A \in \Sigma_x$, which is exactly $(F_0)_* \mu_x = \mu_x$. This further implies

$$w_0 := \frac{d(F_0)_* \mu_x}{d\mu_x} \equiv 1 \quad \mu_x\text{-a.e.}$$

Then by the change-of-variables formula and $(F_0)_* \mu_x = \mu_x$:

$$\begin{aligned} \|C_{F_0} \varphi\|_{\mathcal{H}_x}^2 &= \int_X |(\varphi \circ F_0)(x)|^2 d\mu_x(x) \\ &= \int_X |\varphi(x')|^2 d[(F_0)_* \mu_x](x') \\ &= \int_X |\varphi(x')|^2 d\mu_x(x') = \|\varphi\|_{\mathcal{H}_x}^2. \end{aligned}$$

Thus C_{F_0} is an *isometry* with $\|C_{F_0}\|_{\mathcal{H}_x \rightarrow \mathcal{H}_x} = 1$. If additionally F_0 is bijective and measure-preserving, then $(F_0^{-1})_* \mu_x = \mu_x$ holds automatically and C_{F_0} is *unitary*, recovering the classical setting of [3], which was also employed in [9], [28]. Theorem III.1 generalizes this by allowing for $X \times U$ as the domain of F and deriving a *non-measure preserving* bounded composition operator. The steps taken in relaxing the operator conditions are summarized in the following in Table I, where $\nu_0 := (F_0)_* \mu_x$ and $w_0 := d\nu_0/d\mu_x$.

TABLE I: Comparison of Koopman Operator Settings

Condition on F	R.-N. deriv.	Operator property
$F_0 : X \rightarrow X$, meas.-pres., bijective	$w_0 \equiv 1$	$C_{F_0} : \mathcal{H}_x \rightarrow \mathcal{H}_x$ unitary, $\ C_{F_0}\ = 1$
$F_0 : X \rightarrow X$, meas.-pres., non-bijective	$w_0 \equiv 1$	$C_{F_0} : \mathcal{H}_x \rightarrow \mathcal{H}_x$ isometry, $\ C_{F_0}\ = 1$
$F_0 : X \rightarrow X$, non-meas.-pres., $w_0 \in L^\infty$	$0 \leq w_0 \leq M$	$C_{F_0} : \mathcal{H}_x \rightarrow \mathcal{H}_x$ bounded, $\ C_{F_0}\ \leq M^{1/2}$
$F : X \times U \rightarrow X$, $w \in L^\infty$	$0 \leq w \leq M$	$C_F : \mathcal{H}_x \rightarrow \mathcal{H}$ bounded, $\ C_F\ \leq M^{1/2}$

Note that the use of weighted L^p spaces to establish boundedness of the composition operator for autonomous systems via the Radon-Nikodym derivative was previously employed in [29], in the context of operator Lyapunov equations.

Remark III.3 (Implications for F) Assumptions II.2–II.3 hold whenever $J_F(x, u) := |\det(\partial F/\partial x)(x, u)| \geq c > 0$ a.e. on $X \times U$ for some constant $c > 0$, so that $F_u := x \mapsto F(x, u)$ is a diffeomorphism on X for each fixed $u \in U$, with inverse F_u^{-1} . Taking μ_x as the Lebesgue measure on $X \subseteq \mathbb{R}^n$

and applying Fubini's theorem and the diffeomorphism change-of-variables formula [24, Thm. 7.26] gives

$$\begin{aligned} \nu(A) &= \int_U \mu_x(\{x : F_u(x) \in A\}) d\mu_u(u) \\ &= \int_U \int_A \frac{d\mu_x(x')}{J_F(F_u^{-1}(x'), u)} d\mu_u(u) \leq \frac{\mu_u(U)}{c} \mu_x(A), \end{aligned}$$

for all $A \in \Sigma_x$ ($\mu_x(A) := \int_A d\mu_x$ and similarly for $\mu_u(U)$), where the first equality applies Fubini, the second applies the change of variables $x' = F_u(x)$, and the inequality uses $J_F \geq c$. It follows immediately that $\mu_x(A) = 0 \Rightarrow \nu(A) = 0$, i.e. $\nu \ll \mu_x$ (Assumption II.2). By the Radon-Nikodym theorem [24, Theorem 6.10], there exists $w = d\nu/d\mu_x \in L^1(X, \mu_x)$ such that $\nu(A) = \int_A w d\mu_x$ for all $A \in \Sigma_x$, and the above inequality implies $w(x') \leq c^{-1} \mu_u(U)$ μ_x -a.e., hence $M := \|w\|_{L^\infty} \leq c^{-1} \mu_u(U) < \infty$ (Assumption II.3). The condition $J_F \geq c > 0$ is verifiable. For example, for an Euler discretisation, i.e., $F(x, u) := x + T_s f(x, u)$ with $f(x, u)$ the continuous-time system map, one has $J_F = |\det(I + T_s \partial f/\partial x)|$, and if $\|\partial f/\partial x\|_{\text{op}} \leq L$ uniformly on $X \times U$ with $T_s < 1/L$, then by the bound $|\det(I + T_s Q)| \geq (1 - T_s \|Q\|_{\text{op}})^n$ one obtains $J_F \geq (1 - LT_s)^n =: c > 0$. Here $\|\cdot\|_{\text{op}}$ denotes the matrix operator norm (largest singular value). In the classical autonomous case, F_0 is measure-preserving, which corresponds to $J_F \equiv 1$, a strictly stronger condition than $J_F \geq c > 0$. The latter permits bounded volume distortion but excludes volume collapse, and is sufficient to guarantee that C_F is a bounded operator as shown in Theorem III.1.

A. Coordinate representation in Riesz bases

Having established that $C_F : \mathcal{H}_x \rightarrow \mathcal{H}$ is a bounded operator under Assumptions II.2–II.3, next we will construct its coordinate representation in an orthonormal/Riesz basis as an operator $\mathcal{K} : \ell^2 \rightarrow \ell^2$. This will facilitate the derivation of an exact infinite-dimensional bilinear Koopman model for general nonlinear maps F . To this aim, consider the sets of $\mathbb{K}$ -valued observable functions $\{\psi_{i,x}\}_{i=1}^\infty, \{\psi_{j,u}\}_{j=1}^\infty$ with $\psi_{i,x} \in \mathcal{H}_x, \psi_{j,u} \in \mathcal{H}_u$ for all $i, j \in \mathbb{N}$. Furthermore, we construct lifted state and input vectors, i.e., $\Psi_x(x_t)$, and $\Psi_u(u_t)$, by stacking the corresponding observable functions, e.g., $\Psi_x(x_t) = \text{col}(\psi_{1,x}(x_t), \psi_{2,x}(x_t), \dots)$.

Next, define $\Psi(x, u) := \Psi_x(x) \otimes \Psi_u(u)$ with $\Psi(x, u) = \text{col}(\psi_1(x, u), \psi_2(x, u), \dots)$ and

$$\{\psi_1, \psi_2, \dots\} := \{\psi_{1,x}\psi_{1,u}, \psi_{1,x}\psi_{2,u}, \dots, \psi_{i,x}\psi_{j,u}, \dots\}, \quad (12)$$

i.e., $\{\psi_i\}_{i=1}^\infty$ contains products $\{\psi_{i,x}\psi_{j,u}\}$, $i, j \in \mathbb{N}$, which is a basis of the tensor product $\mathcal{H}_x \otimes \mathcal{H}_u$ cf. Proposition II.5.

Proposition III.4 Let $\{\varphi_i\}_{i=1}^\infty$ and $\{\psi_{i,x}\}_{i=1}^\infty$ be orthonormal bases in $\mathcal{H}_x$ and let $\{\psi_{i,u}\}_{i=1}^\infty$ be an orthonormal basis in $\mathcal{H}_u$. Let the state transition map $F : X \times U \rightarrow X$ in (9) satisfy:

$$\varphi_i \circ F \in \mathcal{H} = \mathcal{H}_x \otimes \mathcal{H}_u, \quad i \in \mathbb{N}. \quad (13)$$

Then there exists an infinite-dimensional matrix $\mathcal{K}$ such that

$$\Psi_x[F(x, u)] = \mathcal{K} \Psi(x, u) = \mathcal{K} (\Psi_x(x) \otimes \Psi_u(u)), \quad (14)$$

with $\Psi_x[F(x, u)] = \text{col}(\psi_{1,x}[F(x, u)], \psi_{2,x}[F(x, u)], \dots)$ and

$$\mathcal{K} = \sum_{j=1}^{\infty} \begin{bmatrix} \langle \psi_{1,x}, \varphi_j \rangle_{\mathcal{H}_x} \\ \langle \psi_{2,x}, \varphi_j \rangle_{\mathcal{H}_x} \\ \vdots \end{bmatrix} \begin{bmatrix} \langle \varphi_j \circ F, \psi_1 \rangle_{\mathcal{H}} & \langle \varphi_j \circ F, \psi_2 \rangle_{\mathcal{H}} & \dots \end{bmatrix}.$$

Proof: Since each $\psi_{i,x} \in \mathcal{H}_x$, $i \in \mathbb{N}$, we can expand each $\psi_{i,x}$ as $\psi_{i,x} = \sum_{j=1}^{\infty} \langle \psi_{i,x}, \varphi_j \rangle_{\mathcal{H}_x} \varphi_j$. Then we have that

$$\begin{aligned} \Psi_x \circ F &= \begin{bmatrix} \psi_{1,x} \circ F \\ \psi_{2,x} \circ F \\ \vdots \end{bmatrix} = \begin{bmatrix} \sum_{j=1}^{\infty} \langle \psi_{1,x}, \varphi_j \rangle_{\mathcal{H}_x} \varphi_j \circ F \\ \sum_{j=1}^{\infty} \langle \psi_{2,x}, \varphi_j \rangle_{\mathcal{H}_x} \varphi_j \circ F \\ \vdots \end{bmatrix} \\ &= \sum_{j=1}^{\infty} \begin{bmatrix} \langle \psi_{1,x}, \varphi_j \rangle_{\mathcal{H}_x} \\ \langle \psi_{2,x}, \varphi_j \rangle_{\mathcal{H}_x} \\ \vdots \end{bmatrix} \varphi_j \circ F. \end{aligned} \quad (15)$$

The second equality holds row-by-row, i.e., for each fixed $i \in \mathbb{N}$, expanding $\psi_{i,x}$ in the ONB $\{\varphi_j\}_{j=1}^{\infty}$ and passing C_F through the sum is justified by boundedness of C_F (Theorem III.1), giving $\psi_{i,x} \circ F = \sum_{j=1}^{\infty} \langle \psi_{i,x}, \varphi_j \rangle_{\mathcal{H}_x} \varphi_j \circ F$ converging in $\mathcal{H}$. The last equality is a purely algebraic rearrangement, i.e., since each row already converges in $\mathcal{H}$, the summation and vector stacking can be interchanged, grouping terms by j rather than by i .

Next, by Proposition II.5 we have that $\{\psi_i\}_{i=1}^{\infty}$ (with ψ_i as defined in (12)) is an orthonormal basis in $\mathcal{H} = \mathcal{H}_x \otimes \mathcal{H}_u$. Then, since $\varphi_j \circ F \in \mathcal{H} = \mathcal{H}_x \otimes \mathcal{H}_u$ for $j \in \mathbb{N}$ it holds that

$$\varphi_j \circ F = \sum_{i=1}^{\infty} \langle \varphi_j \circ F, \psi_i \rangle_{\mathcal{H}} \psi_i, \quad (16)$$

where the inner product on $\mathcal{H}$ is defined as in (4). Substituting (16) in (15) yields

$$\begin{aligned} \Psi_x[F(x, u)] &= \sum_{j=1}^{\infty} \begin{bmatrix} \langle \psi_{1,x}, \varphi_j \rangle_{\mathcal{H}_x} \\ \langle \psi_{2,x}, \varphi_j \rangle_{\mathcal{H}_x} \\ \vdots \end{bmatrix} \varphi_j \circ F(x, u) \\ &= \sum_{j=1}^{\infty} \begin{bmatrix} \langle \psi_{1,x}, \varphi_j \rangle_{\mathcal{H}_x} \\ \langle \psi_{2,x}, \varphi_j \rangle_{\mathcal{H}_x} \\ \vdots \end{bmatrix} \sum_{i=1}^{\infty} \langle \varphi_j \circ F, \psi_i \rangle_{\mathcal{H}} \psi_i(x, u) = \\ &= \sum_{j=1}^{\infty} \begin{bmatrix} \langle \psi_{1,x}, \varphi_j \rangle_{\mathcal{H}_x} \\ \langle \psi_{2,x}, \varphi_j \rangle_{\mathcal{H}_x} \\ \vdots \end{bmatrix} [\langle \varphi_j \circ F, \psi_1 \rangle_{\mathcal{H}}, \langle \varphi_j \circ F, \psi_2 \rangle_{\mathcal{H}}, \dots] \cdot \\ &\quad \cdot \begin{bmatrix} \psi_1(x, u) \\ \psi_2(x, u) \\ \vdots \end{bmatrix} = \mathcal{K} \Psi(x, u), \end{aligned}$$

which completes the proof. ■

The above result generalizes [8, Proposition 2] to state transition maps $F(x, u)$ with control inputs under the generalized invariance property (13); sufficient conditions on F that guarantee this property have already been identified in Theorem III.1. By letting $z_t := \Psi_x(x_t)$ and $v_t := \Psi_u(u_t)$, we

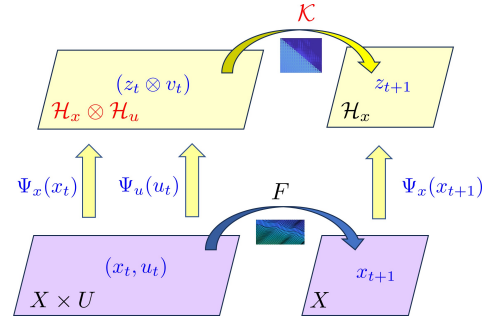


Fig. 2: Illustration of the envisioned generalized Koopman operator architecture for systems with control input.

can define a Koopman infinite-dimensional bilinear system:

$$\begin{aligned} z_{t+1} &= \Psi_x(x_{t+1}) = \Psi_x[F(x_t, u_t)] = \mathcal{K}(\Psi_x(x_t) \otimes \Psi_u(u_t)) \\ &= \mathcal{K}(z_t \otimes v_t), \end{aligned} \quad (17)$$

where $\{x_t, u_t\}_{t \in \mathbb{N}}$ is a solution of system (9). The developed construction of the bilinear Koopman model for systems with inputs is graphically illustrated in Figure 2.

As shown in [8, Corollary 1] for the case of autonomous state transition maps, if in Proposition III.4 we use $\psi_{i,x} = \varphi_i$, $i \in \mathbb{N}$, the infinite-dimensional matrix $\mathcal{K}$ can be simplified to

$$\bar{\mathcal{K}} = \begin{bmatrix} \langle \varphi_1 \circ F, \psi_1 \rangle_{\mathcal{H}} & \langle \varphi_1 \circ F, \psi_2 \rangle_{\mathcal{H}} & \dots \\ \langle \varphi_2 \circ F, \psi_1 \rangle_{\mathcal{H}} & \langle \varphi_2 \circ F, \psi_2 \rangle_{\mathcal{H}} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix}, \quad (18)$$

with $\psi_i(x, u) = \varphi_i(x) \psi_{i,u}(u)$ for all $i \in \mathbb{N}$ for some $j, l \in \mathbb{N}$.

Next, we show that the condition on the sets of observable functions $\{\psi_{i,x}\}_{i=1}^{\infty}$, $\{\psi_{i,u}\}_{i=1}^{\infty}$ can be relaxed to a linearly independent and complete set of functions that form a Riesz basis in $\mathcal{H}_x$ and $\mathcal{H}_u$, respectively, instead of an orthonormal basis. First, notice that we can expand $\psi_{i,x} = \sum_{j=1}^{\infty} \langle \psi_{i,x}, \varphi_j \rangle_{\mathcal{H}_x} \varphi_j$, which can be compactly written as

$$\begin{bmatrix} \psi_{1,x} \\ \psi_{2,x} \\ \vdots \end{bmatrix} = \underbrace{\begin{bmatrix} \langle \psi_{1,x}, \varphi_1 \rangle_{\mathcal{H}_x} & \langle \psi_{1,x}, \varphi_2 \rangle_{\mathcal{H}_x} & \dots \\ \langle \psi_{2,x}, \varphi_1 \rangle_{\mathcal{H}_x} & \langle \psi_{2,x}, \varphi_2 \rangle_{\mathcal{H}_x} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix}}_{:=T_x} \begin{bmatrix} \varphi_1 \\ \varphi_2 \\ \vdots \end{bmatrix}. \quad (19)$$

The same transformation can be done for the product space $\mathcal{H}$, i.e., $\psi_i(x, u) = \sum_{j=1}^{\infty} \langle \psi_i, e_j \rangle_{\mathcal{H}} e_j(x, u)$, which can be compactly written as

$$\begin{bmatrix} \psi_1 \\ \psi_2 \\ \vdots \end{bmatrix} = \underbrace{\begin{bmatrix} \langle \psi_1, e_1 \rangle_{\mathcal{H}} & \langle \psi_1, e_2 \rangle_{\mathcal{H}} & \dots \\ \langle \psi_2, e_1 \rangle_{\mathcal{H}} & \langle \psi_2, e_2 \rangle_{\mathcal{H}} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix}}_{:=T} \begin{bmatrix} e_1 \\ e_2 \\ \vdots \end{bmatrix}, \quad (20)$$

where $\{e_i\}_{i=1}^{\infty}$ is a generic ONB in $\mathcal{H}$. Note that these transformations amount to a change of Riesz basis, since orthonormal bases are Riesz bases [25, Chapter 3.6], and thus, the infinite-dimensional matrices T_x, T are nonsingular, i.e., they are invertible and bounded operators [25, Theorem 3.6.6].

Corollary III.5 *If the observables $\{\psi_{i,x}\}_{i=1}^\infty$ and $\{\psi_{i,u}\}_{i=1}^\infty$ form a Riesz basis in $\mathcal{H}_x$ and $\mathcal{H}_u$, respectively, then the infinite-dimensional matrix $\mathcal{K}$ in (14) is given by*

$$\mathcal{K} = T_x \bar{\mathcal{K}} T^{-1}, \quad (21)$$

with $\bar{\mathcal{K}}$, T_x and T defined as in (18)-(20), respectively.

So far, we have shown the existence of the matrix $\mathcal{K}$, but its expression depends on a generic, ONB $\{\varphi_i\}_{i=1}^\infty$ in $\mathcal{H}_x$. Next, we prove that $\mathcal{K}$ can be constructed just based on the state and input observable functions and the map F .

Theorem III.6 *Let $\{\psi_{i,x}\}_{i=1}^\infty$ and $\{\psi_{i,u}\}_{i=1}^\infty$ be independent and complete sets of Riesz basis functions spanning $\mathcal{H}_x$, and $\mathcal{H}_u$, respectively. Let the state transition map $F : X \times U \rightarrow X$ in (9) satisfy:*

$$\psi_{i,x} \circ F \in \mathcal{H}, \quad i \in \mathbb{N}. \quad (22)$$

Then there exists an infinite-dimensional matrix $\mathcal{K} = Q_x R^{-1}$ such that

$$\Psi_x(F(x, u)) = \mathcal{K} \Psi(x, u) = \mathcal{K}(\Psi_x(x) \otimes \Psi_u(u)), \quad (23)$$

with the matrices R and Q_x given by:

$$R = \begin{bmatrix} \langle \psi_1, \psi_1 \rangle_{\mathcal{H}} & \langle \psi_1, \psi_2 \rangle_{\mathcal{H}} & \dots \\ \langle \psi_2, \psi_1 \rangle_{\mathcal{H}} & \langle \psi_2, \psi_2 \rangle_{\mathcal{H}} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix} \quad (24)$$

and

$$Q_x = \begin{bmatrix} \langle \psi_{1,x} \circ F, \psi_1 \rangle_{\mathcal{H}} & \langle \psi_{1,x} \circ F, \psi_2 \rangle_{\mathcal{H}} & \dots \\ \langle \psi_{2,x} \circ F, \psi_1 \rangle_{\mathcal{H}} & \langle \psi_{2,x} \circ F, \psi_2 \rangle_{\mathcal{H}} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix}. \quad (25)$$

Proof: From (19) we obtain

$$\begin{bmatrix} \varphi_1 \circ F \\ \varphi_2 \circ F \\ \vdots \end{bmatrix} = T_x^{-1} \begin{bmatrix} \psi_{1,x} \circ F \\ \psi_{2,x} \circ F \\ \vdots \end{bmatrix},$$

which, together with (22) implies $\varphi_i \circ F \in \mathcal{H}$ for all $i \in \mathbb{N}$. Then, from Proposition III.4 we have that there exists an infinite-dimensional matrix $\mathcal{K}$ such that (23) holds. Multiplying both sides of the equation (23) on the right by the row vector $[\psi_1, \psi_2, \dots]$ gives the matrix of functions equality

$$\bar{Q}(x, u) = \mathcal{K} \bar{R}(x, u) \quad (26)$$

where $\{\psi_j\}_{j=1}^\infty$ is the product Riesz basis of $\mathcal{H} = \mathcal{H}_x \otimes \mathcal{H}_u$ and

$$\begin{aligned} [\bar{Q}(x, u)]_{ij} &:= \psi_{i,x}(F(x, u)) \cdot \psi_j(x, u), \\ [\bar{R}(x, u)]_{ij} &:= \psi_i(x, u) \cdot \psi_j(x, u). \end{aligned}$$

For each fixed $i, j \in \mathbb{N}$, integrating the (i, j) -th entry of both sides of (26) over $X \times U$ gives on the left:

$$\begin{aligned} & \int_{X \times U} [\bar{Q}(x, u)]_{ij} d(\mu_x \otimes \mu_u) \\ &= \int_{X \times U} \psi_{i,x}(F(x, u)) \psi_j(x, u) d(\mu_x \otimes \mu_u) \\ &= \langle \psi_{i,x} \circ F, \psi_j \rangle_{\mathcal{H}} = [Q_x]_{ij}, \end{aligned}$$

and on the right:

$$\begin{aligned} & \int_{X \times U} [\mathcal{K} \bar{R}(x, u)]_{ij} d(\mu_x \otimes \mu_u) \\ &= \int_{X \times U} \sum_{k=1}^{\infty} [\mathcal{K}]_{ik} \psi_k(x, u) \psi_j(x, u) d(\mu_x \otimes \mu_u). \end{aligned}$$

Define the sums $f_N(x, u) := \sum_{k=1}^N [\mathcal{K}]_{ik} \psi_k(x, u) \psi_j(x, u)$, so that the infinite sum is the pointwise limit $\lim_{N \rightarrow \infty} f_N(x, u) = \sum_{k=1}^{\infty} [\mathcal{K}]_{ik} \psi_k(x, u) \psi_j(x, u)$. Next we establish that

$$\int_{X \times U} \lim_{N \rightarrow \infty} f_N d(\mu_x \otimes \mu_u) = \lim_{N \rightarrow \infty} \int_{X \times U} f_N d(\mu_x \otimes \mu_u). \quad (27)$$

This holds by the dominated convergence theorem [24, Theorem 1.34] if there exists $g \in L^1(X \times U, \mu_x \otimes \mu_u)$ such that $|f_N(x, u)| \leq g(x, u)$ for all N . Such a dominating function exists as follows: Cauchy-Schwarz pointwise gives

$$\begin{aligned} |f_N(x, u)| &\leq \left(\sum_{k=1}^N |[\mathcal{K}]_{ik}|^2 \right)^{1/2} \left(\sum_{k=1}^N |\psi_k(x, u)|^2 \right)^{1/2} |\psi_j(x, u)| \\ &\leq \|\mathcal{K}\| \left(\sum_{k=1}^{\infty} |\psi_k(x, u)|^2 \right)^{1/2} |\psi_j(x, u)| =: g(x, u), \end{aligned}$$

uniformly in N , where the second inequality bounds the finite row norm by $\|\mathcal{K}\|$ and extends the sum to infinity. Then the Bessel upper bound gives $\sum_k |\psi_k(x, u)|^2 \leq \beta$ a.e., so $g(x, u) \leq \|\mathcal{K}\| \beta^{1/2} |\psi_j(x, u)|$ and hence $\int g d(\mu_x \otimes \mu_u) \leq \|\mathcal{K}\| \beta^{1/2} (\mu_x(X) \mu_u(U))^{1/2} \|\psi_j\|_{\mathcal{H}} < \infty$, where we used

$$\begin{aligned} \int_{X \times U} |\psi_j| d(\mu_x \otimes \mu_u) &\leq \|1\|_{\mathcal{H}} \|\psi_j\|_{\mathcal{H}} \\ &= (\mu_x(X) \mu_u(U))^{1/2} \|\psi_j\|_{\mathcal{H}} < \infty. \end{aligned}$$

Thus $g \in L^1$ and (27) holds by the dominated convergence theorem [24, Theorem 1.34]. Hence, it follows that

$$\begin{aligned} & \int_{X \times U} [\mathcal{K} \bar{R}(x, u)]_{ij} d(\mu_x \otimes \mu_u) \\ &= \int_{X \times U} \sum_{k=1}^{\infty} [\mathcal{K}]_{ik} \psi_k(x, u) \psi_j(x, u) d(\mu_x \otimes \mu_u) \\ &= \sum_{k=1}^{\infty} [\mathcal{K}]_{ik} \underbrace{\int_{X \times U} \psi_k(x, u) \psi_j(x, u) d(\mu_x \otimes \mu_u)}_{= \langle \psi_k, \psi_j \rangle_{\mathcal{H}} = [R]_{kj}} = [\mathcal{K} R]_{ij}. \end{aligned}$$

Since $[Q_x]_{ij} = [\mathcal{K} R]_{ij}$ holds for all $i, j \in \mathbb{N}$, we conclude $Q_x = \mathcal{K} R$. Since $\{\psi_i\}_{i=1}^\infty$ is a Riesz basis in $\mathcal{H} = \mathcal{H}_x \otimes \mathcal{H}_u$, its Gram matrix R is nonsingular/invertible [25, Theorem 3.6.6], which yields $\mathcal{K} = Q_x R^{-1}$. ■

Remark III.7 (Boundedness of $\mathcal{K}$ and its relation to C_F)
The infinite-dimensional matrix $\mathcal{K} = Q_x R^{-1}$ defines a bounded linear operator $\mathcal{K} : \ell^2 \rightarrow \ell^2$. Indeed,

$$\mathcal{K} = Q_x R^{-1} = S^* \circ C_F \circ S_x \circ R^{-1}, \quad (28)$$

where $S_x : \ell^2 \rightarrow \mathcal{H}_x$ and $S : \ell^2 \rightarrow \mathcal{H}$ are the synthesis operators [25, Ch. 3] of the Riesz bases in $\mathcal{H}_x$ and $\mathcal{H}$, respectively,

with adjoints $S_x^* : \mathcal{H}_x \rightarrow \ell^2$ and $S^* : \mathcal{H} \rightarrow \ell^2$ the corresponding analysis operators. Above, R is the Gram matrix of the product Riesz basis in $\mathcal{H}$ with $[R]_{i,j} = \langle \psi_i(x, u), \psi_j(x, u) \rangle_{\mathcal{H}}$. The lifted dynamics $z_{t+1} = \mathcal{K}(z_t \otimes v_t)$ are the coordinate-space expression of $\psi_{i,x}(x_{t+1}) = \psi_{i,x}(F(x_t, u_t))$ for all $i \in \mathbb{N}$, connected to C_F via the Riesz basis isomorphism. Boundedness of $\mathcal{K}$ follows from the factorization $\mathcal{K} = Q_x R^{-1}$ in (28). The factor $R^{-1} : \ell^2 \rightarrow \ell^2$ is bounded with $\|R^{-1}\| \leq \alpha^{-1}$, since the Riesz basis lower bound gives $R \geq \alpha I$ on ℓ^2 [25, Theorem 3.6.6]. $Q_x : \ell^2 \rightarrow \ell^2$ is bounded since

$$\|Q_x\| \leq \|S^*\| \|C_F\|_{\mathcal{H}_x \rightarrow \mathcal{H}} \|S_x\| \leq \beta M^{1/2}, \quad (29)$$

where $\|S_x\|, \|S^*\| \leq \beta^{1/2}$ by the Bessel property of the Riesz bases [25, Theorem 3.2.3] and $\|C_F\|_{\mathcal{H}_x \rightarrow \mathcal{H}} \leq M^{1/2}$ by Theorem III.1. Submultiplicativity then gives

$$\|\mathcal{K}\|_{\ell^2 \rightarrow \ell^2} \leq \frac{M^{1/2} \beta}{\alpha}, \quad (30)$$

where β/α is the Riesz basis condition number.

B. Input-affine nonlinear systems

Consider the input-affine discrete-time system

$$x_{t+1} = F(x_t, u_t) := f(x_t) + \sum_{j=1}^m g_j(x_t) u_{j,t}, \quad (31)$$

with $f, g_j : X \rightarrow \mathbb{R}^n$ measurable, $u = (u_1, \dots, u_m)^\top \in U \subseteq \mathbb{R}^m$.

Definition III.8 Let the $m+1$ scalar-valued input observables $\psi_{j,u} : U \rightarrow \mathbb{R}$ be defined as

$$\psi_{0,u}(u) := 1, \quad \psi_{j,u}(u) := u_j, \quad j = 1, \dots, m. \quad (32)$$

The corresponding input observable subspace is

$$\mathcal{V}_u := \text{span}\{\psi_{0,u}, \psi_{1,u}, \dots, \psi_{m,u}\} \subset L^2(U, \mu_u), \quad (33)$$

which has dimension $m+1$ since $\{\psi_{j,u}\}_{j=0}^m$ are linearly independent in $L^2(U, \mu_u)$ for any $U \subseteq \mathbb{R}^m$ of positive Lebesgue measure. Define $\mathcal{H}_u := \mathcal{V}_u$ and the map

$$\begin{aligned} \Psi_u(u_t) &:= v_t := (\psi_{0,u}(u_t), \psi_{1,u}(u_t), \dots, \psi_{m,u}(u_t))^\top \\ &= (1, u_{1,t}, \dots, u_{m,t})^\top \in \mathbb{R}^{m+1}. \end{aligned} \quad (34)$$

Assumption III.9 For each $i \in \mathbb{N}$, $\psi_{i,x} \circ F \in \mathcal{H}_x \otimes \mathcal{V}_u$, i.e., there exist $a_i, b_{ij} \in \mathcal{H}_x$ such that

$$\psi_{i,x}(F(x, u)) = a_i(x) \psi_{0,u}(u) + \sum_{j=1}^m b_{ij}(x) \psi_{j,u}(u). \quad (35)$$

Theorem III.10 Let $\{\psi_{i,x}\}_{i=1}^\infty$ be a Riesz basis for $\mathcal{H}_x$, let $\{\psi_{j,u}\}_{j=0}^m$ be as in Definition III.8, and let Assumption III.9 hold. Let $\{e_j\}_{j=0}^m$ denote the standard basis of $\mathbb{R}^{m+1}$. Then there exist bounded operators $\mathcal{K} : \ell^2 \rightarrow \ell^2$, $\mathcal{A} : \ell^2 \rightarrow \ell^2$ and $\mathcal{N}_j : \ell^2 \rightarrow \ell^2$, $j = 1, \dots, m$, such that

$$z_{t+1} = \mathcal{K}(z_t \otimes v_t) = \mathcal{A}z_t + \sum_{j=1}^m \mathcal{N}_j z_t u_{j,t}, \quad (36)$$

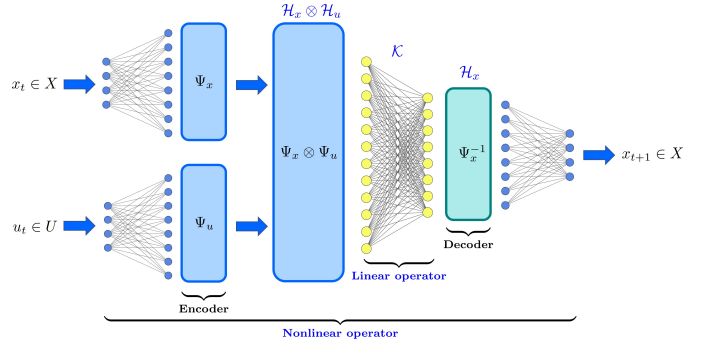


Fig. 3: Generalized Koopman operator learning architecture: (i) State and input at time t are lifted via Encoders; (ii) Tensor product of the resulting lifted state and input vectors; (iii) Koopman linear operator advances the lifted state to $t+1$; (iv) State at time $t+1$ is retrieved using a Decoder.

$$\text{with } \mathcal{K} = \mathcal{A} \otimes e_0^\top + \sum_{j=1}^m \mathcal{N}_j \otimes e_j^\top.$$

The proof is given in Appendix I.

Remark III.11 (Relation with exact bilinearization)

Assumption III.9 can be regarded as the discrete-time counterpart of the necessary and sufficient conditions of [12] for exact bilinearization of continuous-time control-affine systems, requiring the autonomous part and input coefficient functions of each observable to lie in $\mathcal{H}_x$. In [11], these conditions are enforced via closure under Lie derivatives along the input vector fields. The linear-in-control model of [9], $z_{t+1} = Az_t + Bu_t$, corresponds to the special case $b_{ij}(x) = [B]_{ij}$ constant, which is generally not satisfied when $g_j(x)$ is state-dependent.

C. Data-driven finite-dimensional approximation

In this section we provide brief guidelines for applying the EDMD approach [7] to the developed generalized Koopman operator. The main idea of EDMD in its simplest form, i.e., least-squares regression, is to generate a set of data snapshots $\{(x_t, u_t, x_{t+1})\}_{t \in \mathbb{N}_{[0, T-1]}}$, $T \in \mathbb{N}_{\geq 1}$ using the nonlinear dynamical system (9), i.e., such that it holds that $x_{t+1} = F(x_t, u_t)$ for all t . Then, letting $z_t = \Psi_x(x_t) \in \mathbb{R}^{n_z}$, $v_t = \Psi_u(u_t) \in \mathbb{R}^{n_v}$, for any $t \in \mathbb{N}$, where the vectors of observables are now finite-dimensional, we can define the data matrices:

$$\begin{aligned} Z_X &= [\Psi_x(x_0) \quad \Psi_x(x_1) \quad \dots \quad \Psi_x(x_{T-1})] \\ &= [z_0 \quad z_1 \quad \dots \quad z_{T-1}] \in \mathbb{R}^{n_z \times T}, \end{aligned} \quad (37a)$$

$$\begin{aligned} V_U &= [\Psi_u(u_0) \quad \Psi_u(u_1) \quad \dots \quad \Psi_u(u_{T-1})] \\ &= [v_0 \quad v_1 \quad \dots \quad v_{T-1}] \in \mathbb{R}^{n_v \times T}, \end{aligned} \quad (37b)$$

$$\begin{aligned} Z_X^+ &= [\Psi_x(x_1) \quad \Psi_x(x_2) \quad \dots \quad \Psi_x(x_T)] \\ &= [z_1 \quad z_2 \quad \dots \quad z_T] \in \mathbb{R}^{n_z \times T}. \end{aligned} \quad (37c)$$

Above $n_z, n_v \in \mathbb{N}$ correspond to the number of state and input observable functions, respectively. Next, differently from standard EDMD for systems with control input, which computes

matrices (A, B) such that $Z_X^+ \approx AZ_X + BV_U$, our aim is to compute a matrix $\hat{\mathcal{K}} \in \mathbb{R}^{n_z \times n_z n_v}$ such that

$$\begin{aligned} Z_X^+ &= \hat{\mathcal{K}}(Z_X \odot V_U) \\ &= \hat{\mathcal{K}} \begin{bmatrix} z_0 \otimes v_0 & z_1 \otimes v_1 & \dots & z_{T-1} \otimes v_{T-1} \end{bmatrix}. \end{aligned} \quad (38)$$

Above $\odot$ denotes the Khatri-Rao product and $\otimes$ denotes the Kronecker product of two columns/vectors. Assuming that the matrix $(Z_X \odot V_U)$ has full-row rank, we can compute a finite-dimensional approximation of $\mathcal{K}$ as

$$\hat{\mathcal{K}} := Z_X^+(Z_X \odot V_U)^\dagger. \quad (39)$$

Using a randomised SVD [30] applied to the Khatri-Rao matrix $(Z_X \odot V_U) \in \mathbb{R}^{n_z n_v \times T}$, the computational cost reduces to $O(n_z n_v \cdot T \cdot r)$, where $r \ll n_z n_v$ is the target rank, making the method scalable to large datasets via GPU implementation.

Figure 3 illustrates the developed architecture for learning generalized Koopman operators, in which kernel functions, deep neural networks or Takens delay embeddings can be used as observable functions.

Remark III.12 (Convergence analysis) Truncating to n_z state and n_v input observables, the total EDMD approximation error decomposes as

$$\begin{aligned} \|\mathcal{K} - \hat{\mathcal{K}}^{(n_z, n_v)}\| &\leq \underbrace{\|\mathcal{K} - \mathcal{K}^{(n_z, n_v)}\|}_{\text{Galerkin error}} + \underbrace{\|\mathcal{K}^{(n_z, n_v)} - \hat{\mathcal{K}}^{(n_z, n_v)}\|}_{\text{data error}}, \end{aligned} \quad (40)$$

where $\mathcal{K}^{(n_z, n_v)}$ is the exact Galerkin projection onto the truncated subspace. The Galerkin error vanishes as $n_z, n_v \rightarrow \infty$ by completeness of the Riesz bases and boundedness of $\mathcal{K}$ (Remark III.7), provided the truncated subspaces are nested with dense union in ℓ^2 ; the analogous result for orthonormal bases is proved in [28]. When the observables are universal kernels [31] on compact X, U and F is sufficiently smooth, the functions $\psi_{i,x} \circ F$ lie in the product RKHS $\mathcal{H}_{k_x} \otimes \mathcal{H}_{k_u}$, which is dense in $C(X \times U)$ by universality; if additionally $\mu_x \otimes \mu_u$ is a regular Borel measure, then $C(X \times U)$ is dense in $L^2(X \times U, \mu_x \otimes \mu_u)$ [24, Theorem 3.14], and the Galerkin error decays at the kernel interpolation rate [32]. The data error is $O(T^{-1/2})$ by concentration inequalities for empirical inner products when the data triples are independently sampled [33].

Remark III.13 Finite-dimensional EDMD approximation with n_z state and n_v input observables does not require that the generalized invariance condition $\psi_{i,x} \circ F \in \mathcal{H}$ holds for the truncated system, i.e., the finite matrix $\hat{\mathcal{K}}^{(n_z, n_v)}$ is always well-defined as the Galerkin projection of the infinite-dimensional $\mathcal{K}$ onto the finite subspace $\mathcal{V}_{n_z} \otimes \mathcal{V}_{n_v}$. Boundedness of $\mathcal{K}$ (Remark III.7) and completeness of the Riesz basis guarantee that the Galerkin truncation error converges to 0 as $n_z, n_v \rightarrow \infty$ [28]. Invariance is not required for this convergence, which is consistent with classical EDMD. Alternatively, one can aim to learn a finite-dimensional invariant subspace, as done in [34]. In finite-dimensional EDMD, one can employ the extended

bilinear Koopman model

$$\begin{aligned} z_{t+1} &= Az_t + Bv_t + \hat{\mathcal{K}}(z_t \otimes v_t) = \\ &A \begin{bmatrix} x_t \\ \Psi_x(x_t) \end{bmatrix} + B \begin{bmatrix} u_t \\ \Psi_u(u_t) \end{bmatrix} + \hat{\mathcal{K}} \left(\begin{bmatrix} x_t \\ \Psi_x(x_t) \end{bmatrix} \otimes \begin{bmatrix} u_t \\ \Psi_u(u_t) \end{bmatrix} \right). \end{aligned}$$

Including unlifted states and inputs yields injectivity and it enables cost functions and constraints in the original system variables; the linear terms can also compensate for residuals.

IV. THE NONLINEAR FUNDAMENTAL LEMMA: A BILINEAR-KOOPMAN APPROACH

In order to derive an analogue of the fundamental lemma [4] for nonlinear systems, consider next the nonlinear system (9) with a state-dependent output added, i.e.,

$$\begin{aligned} x_{t+1} &= F(x_t, u_t), \quad t \in \mathbb{N} \\ y_t &= h(x_t), \end{aligned} \quad (41)$$

where $h : X \rightarrow Y$, $Y \subseteq \mathbb{R}^p$ is a suitable mapping. We assume that h is Lebesgue integrable and we introduce the Hilbert space $\mathcal{H}_y = L^2(Y, \mu_y)$ on $Y \subseteq \mathbb{R}^p$ with the standard inner product $\langle \cdot, \cdot \rangle_{\mathcal{H}_y} : \mathcal{H}_y \times \mathcal{H}_y \rightarrow \mathbb{K}$.

A. Koopman representations with lifted outputs

Next, we consider a composition operator $C_h : \mathcal{H}_y \rightarrow \mathcal{H}_x$ defined by $(C_h \varphi)(x) = \varphi(h(x))$ for a measurable output map $h : X \rightarrow Y$ and we directly prove existence of an infinite-dimensional matrix $\mathcal{C}$ that is the coordinate representation of C_h in a Riesz basis, as done for the generalized Koopman operator. To this end we introduce the following set of $\mathbb{K}$ -valued observable functions for the output of system (41), i.e., $\{\psi_{i,y}\}_{i=1}^\infty$, with $\psi_{i,y} \in \mathcal{H}_y$ for all $i \in \mathbb{N}$. Furthermore, we construct a lifted output vector $\Psi_y(y_t)$ by stacking the corresponding observable functions, e.g., $\Psi_y(y_t) = \text{col}(\psi_{1,y}(y_t), \psi_{2,y}(y_t), \dots)$.

Proposition IV.1 Let $\{\psi_{i,x}\}_{i=1}^\infty$ be an orthonormal basis in $\mathcal{H}_x$ and let $\{\phi_i\}_{i=1}^\infty$ and $\{\psi_{i,y}\}_{i=1}^\infty$ be two orthonormal bases in $\mathcal{H}_y$. Let the output map $h : X \rightarrow Y$ in (41) satisfy:

$$\phi_i \circ h \in \mathcal{H}_x, \quad i \in \mathbb{N}. \quad (42)$$

Then there exists an infinite-dimensional matrix $\mathcal{C}$ such that

$$\Psi_y[h(x)] = \mathcal{C}\Psi_x(x), \quad (43)$$

where

$$\begin{aligned} \mathcal{C} &= \sum_{j=1}^\infty \begin{bmatrix} \langle \psi_{1,y}, \phi_j \rangle_{\mathcal{H}_y} \\ \langle \psi_{2,y}, \phi_j \rangle_{\mathcal{H}_y} \\ \vdots \end{bmatrix} \\ &\quad \cdot [\langle \phi_j \circ h, \psi_{1,x} \rangle_{\mathcal{H}_x} \quad \langle \phi_j \circ h, \psi_{2,x} \rangle_{\mathcal{H}_x} \quad \dots]. \end{aligned}$$

As done for $\mathcal{K}$, if we choose $\psi_{i,y} = \phi_i$, $i \in \mathbb{N}$, the output matrix $\mathcal{C}$ can be simplified to

$$\bar{\mathcal{C}} = \begin{bmatrix} \langle \phi_1 \circ h, \psi_{1,x} \rangle_{\mathcal{H}_x} & \langle \phi_1 \circ h, \psi_{2,x} \rangle_{\mathcal{H}_x} & \dots \\ \langle \phi_2 \circ h, \psi_{1,x} \rangle_{\mathcal{H}_x} & \langle \phi_2 \circ h, \psi_{2,x} \rangle_{\mathcal{H}_x} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix} \quad (44)$$

and the condition on the set of observable functions $\{\psi_{i,y}\}_{i=1}^\infty$ can be relaxed to a linearly independent and complete set of functions that form a Riesz basis in $\mathcal{H}_y$. First, notice that we can expand $\psi_{i,y} = \sum_{j=1}^\infty \langle \psi_{i,y}, \phi_j \rangle_{\mathcal{H}_y} \phi_j$, which can be compactly written as

$$\begin{bmatrix} \psi_{1,y} \\ \psi_{2,y} \\ \vdots \end{bmatrix} = \underbrace{\begin{bmatrix} \langle \psi_{1,y}, \phi_1 \rangle_{\mathcal{H}_y} & \langle \psi_{1,y}, \phi_2 \rangle_{\mathcal{H}_y} & \cdots \\ \langle \psi_{2,y}, \phi_1 \rangle_{\mathcal{H}_y} & \langle \psi_{2,y}, \phi_2 \rangle_{\mathcal{H}_y} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}}_{:=T_y} \begin{bmatrix} \phi_1 \\ \phi_2 \\ \vdots \end{bmatrix}, \quad (45)$$

where the infinite-dimensional matrix T_y is nonsingular.

Corollary IV.2 *If the observables $\{\psi_{i,x}\}_{i=1}^\infty$ and $\{\psi_{i,y}\}_{i=1}^\infty$ form a Riesz basis in $\mathcal{H}_x$ and $\mathcal{H}_y$, respectively, then the infinite-dimensional matrix $\mathcal{C}$ in (43) is given by*

$$\mathcal{C} = T_y \bar{\mathcal{C}} T_x^{-1}, \quad (46)$$

with $T_x, T_y, \bar{\mathcal{C}}$ defined in (19), (45) and (44), respectively.

The version of Theorem III.6 for the output is given next.

Theorem IV.3 *Let $\{\psi_{i,x}\}_{i=1}^\infty$ and $\{\psi_{i,y}\}_{i=1}^\infty$ be independent and complete sets of Riesz basis functions spanning $\mathcal{H}_x$ and $\mathcal{H}_y$, respectively. Let the output map $h : X \rightarrow Y$ in (41) satisfy:*

$$\psi_{i,y} \circ h \in \mathcal{H}_x, \quad i \in \mathbb{N}. \quad (47)$$

Then there exists an infinite-dimensional matrix $\mathcal{C} = Q_y R_x^{-1}$ such that

$$\Psi_y[h(x)] = \mathcal{C} \Psi_x(x), \quad (48)$$

with the matrices R_x and Q_y given by:

$$R_x = \begin{bmatrix} \langle \psi_{1,x}, \psi_{1,x} \rangle_{\mathcal{H}_x} & \langle \psi_{1,x}, \psi_{2,x} \rangle_{\mathcal{H}_x} & \cdots \\ \langle \psi_{2,x}, \psi_{1,x} \rangle_{\mathcal{H}_x} & \langle \psi_{2,x}, \psi_{2,x} \rangle_{\mathcal{H}_x} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} \quad (49)$$

and

$$Q_y = \begin{bmatrix} \langle \psi_{1,y} \circ h, \psi_{1,x} \rangle_{\mathcal{H}_x} & \langle \psi_{1,y} \circ h, \psi_{2,x} \rangle_{\mathcal{H}_x} & \cdots \\ \langle \psi_{2,y} \circ h, \psi_{1,x} \rangle_{\mathcal{H}_x} & \langle \psi_{2,y} \circ h, \psi_{2,x} \rangle_{\mathcal{H}_x} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}. \quad (50)$$

For the proofs of Proposition IV.1 and Theorem IV.3 we refer to Appendix II and Appendix III, respectively.

Remark IV.4 (Boundedness of C_h and local observability)

Analogous conditions to Assumptions II.2–II.3 can be derived to guarantee that the composition operator $C_h : \mathcal{H}_y \rightarrow \mathcal{H}_x$ defined by $(C_h \varphi)(x) = \varphi(h(x))$ for a measurable output map $h : X \rightarrow Y$ is a well-defined bounded linear operator. Specifically, non-singularity and boundedness of the Radon-Nikodym derivative $w_h = dh_* \mu_x / d\mu_y \in L^\infty(Y, \mu_y)$ are sufficient for C_h to be bounded with $\|C_h\|_{\mathcal{H}_y \rightarrow \mathcal{H}_x} \leq \|w_h\|_{L^\infty}^{1/2}$, by the same argument as in Theorem III.1. For C^1 maps $h : X \rightarrow Y$ with $\dim(Y) \leq \dim(X)$, these conditions reduce via the coarea formula [35, Section 3.4] to

$$J_h(x) := \sqrt{\det \left(\frac{\partial h}{\partial x}(x) \left(\frac{\partial h}{\partial x}(x) \right)^\top \right)} \geq c > 0,$$

a.e. on X . This requires that $\partial h / \partial x(x)$ has full row rank $\dim(Y)$ a.e., i.e., distinct states are locally distinguishable through h or equivalently, *uniform local observability* holds. Furthermore, similarly as derived for $\mathcal{K}$, it can be established that the infinite-dimensional matrix $\mathcal{C}$ defines a well-defined bounded linear operator on ℓ^2 .

Let $w_t := \Psi_y(y_t)$ and define the following exact infinite-dimensional bilinear Koopman system with a linear output:

$$\begin{aligned} z_{t+1} &= \mathcal{K}(z_t \otimes v_t), \\ w_t &= \Psi_y(y_t) = \Psi_y[h(x_t)] = \mathcal{C} \Psi_x(x_t) \\ &= \mathcal{C} z_t. \end{aligned} \quad (51)$$

Note that Koopman models with lifted outputs have been previously considered in [36] to study observability properties.

Now we are ready to define a relation between the solutions of the nonlinear system (41) and the infinite-dimensional bilinear Koopman system (51).

Theorem IV.5 *Let $\{\psi_{i,x}\}_{i=1}^\infty$, $\{\psi_{i,u}\}_{i=1}^\infty$ and $\{\psi_{i,y}\}_{i=1}^\infty$ be independent and complete sets of Riesz basis functions spanning $\mathcal{H}_x$, $\mathcal{H}_u$ and $\mathcal{H}_y$, respectively. Let the state transition map $F : X \times U \rightarrow X$ and the output map $h : X \rightarrow Y$ in (9) satisfy:*

$$\begin{aligned} \psi_{i,x} \circ F &\in \mathcal{H}_x, \quad i \in \mathbb{N}, \\ \psi_{i,y} \circ h &\in \mathcal{H}_x, \quad i \in \mathbb{N}. \end{aligned} \quad (52)$$

Let $\{\varphi_i\}_{i=1}^\infty$ and $\{\phi_i\}_{i=1}^\infty$ be orthonormal bases in $\mathcal{H}_x$, $\mathcal{H}_y$, and let $\mathcal{K}$ and $\mathcal{C}$ be defined as in (21), (46). Assume that the state and output lifting mappings $\Psi_x(x) = \text{col}(\psi_{1,x}(x), \psi_{2,x}(x), \dots)$ and $\Psi_y(y) = \text{col}(\psi_{1,y}(y), \psi_{2,y}(y), \dots)$ are injective.

Then $\{x_t, u_t, y_t\}_{t \in \mathbb{N}}$ is a solution of system (41) if and only if $\{z_t, v_t, w_t\}_{t \in \mathbb{N}}$ is a solution of system (51).

Proof: Suppose that $\{x_t, u_t, y_t\}_{t \in \mathbb{N}}$ is a solution of system (41). Then from Proposition III.4-Corollary III.5 and Proposition IV.1-Corollary IV.2 we obtain that $\{z_t, v_t, w_t\}_{t \in \mathbb{N}} = \{\Psi_x(x_t), \Psi_u(u_t), \Psi_y(y_t)\}_{t \in \mathbb{N}}$ is a solution of system (51). Note that this part of the proof does not require that the mappings Ψ_x, Ψ_y are injective.

Conversely, let $\{x_t, u_t, y_t\}_{t \in \mathbb{N}}$ be such that $\{z_t, v_t, w_t\}_{t \in \mathbb{N}}$ with $z_t = \Psi_x(x_t)$, $v_t = \Psi_u(u_t)$ and $w_t = \Psi_y(y_t)$ is a solution of system (51). For all $t \in \mathbb{N}$ let $\hat{x}_{t+1} = F(x_t, u_t)$ and let $\hat{y}_t = h(x_t)$ and suppose that $\hat{x}_{t+1} \neq x_{t+1}$ and $\hat{y}_t \neq y_t$ for some $t \in \mathbb{N}$. Then, by Corollary III.5 and Corollary IV.2 we have that $\Psi_x(\hat{x}_{t+1}) = \mathcal{K} \Psi(x_t, u_t) = \Psi_x(x_{t+1})$ and $\Psi_y(\hat{y}_t) = \mathcal{C} \Psi_x(x_t) = \Psi_y(y_t)$. If the mappings Ψ_x and Ψ_y are injective, then it must hold that $\hat{x}_{t+1} = x_{t+1}$ and $\hat{y}_t = y_t$ and thus, $\{x_t, u_t, y_t\}_{t \in \mathbb{N}}$ is a solution of system (41). ■

B. The nonlinear fundamental lemma

Let $N_d \in \mathbb{N}$ data sets be collected from system (41), each from initial condition $x_0^{(i)}$, $i = 1, \dots, N_d$, with trajectory length $T_i + N$, giving lifted trajectories $z_t^{(i)} = \Psi_x(x_t^{(i)}) \in \ell^2$, $v_t^{(i)} = \Psi_u(u_t^{(i)}) \in \ell^2$, $w_t^{(i)} = \Psi_y(y_t^{(i)}) \in \ell^2$. The N -step lifted sequence from data set i at time s is

$$\Phi_s^{(i)} := \text{col} \left(z_s^{(i)} \otimes v_s^{(i)}, \dots, z_{s+N-1}^{(i)} \otimes v_{s+N-1}^{(i)} \right) \in \ell^2.$$

Definition IV.6 (Multiple-data-sets Hankel operator)

Let $\mathcal{T} := \bigoplus_{i=1}^{N_d} \mathcal{G}_i$ with $\mathcal{G}_i = \mathbb{R}^{T_i}$ for $T_i < \infty$ and $\mathcal{G}_i = \ell^2$ for $T_i = \infty$, with elements $g = (g^{(1)}, \dots, g^{(N_d)})$, $g^{(i)} = (g_0^{(i)}, g_1^{(i)}, \dots)$. Define the *multiple-data-sets Hankel operator* as

$$\mathcal{F}_N : \mathcal{T} \rightarrow \ell^2, \quad \mathcal{F}_N g := \sum_{i=1}^{N_d} \sum_{s=0}^{T_i-1} g_s^{(i)} \Phi_s^{(i)}, \quad (53)$$

where, for $T_i = \infty$, the inner sum converges in ℓ^2 by square-summability of $g^{(i)} \in \ell^2$.

Definition IV.7 The *N-step truncated behavior* of the bilinear Koopman system (51) is defined as the set of all *N-step admissible data sequences* $\Phi_t := \text{col}(z_t \otimes v_t, z_{t+1} \otimes v_{t+1}, \dots, z_{t+N-1} \otimes v_{t+N-1}) \in \ell^2$:

$$\begin{aligned} \mathcal{B}_N := \\ \{ \Phi_t \in \ell^2 \mid z_t \in \ell^2, v_{[t, t+N-1]} \in \ell^2, z_{i+1} = \mathcal{K}(z_i \otimes v_i), \\ i = t, \dots, t+N-1. \} \end{aligned} \quad (54)$$

Let $\mathcal{I} := \{(i, s) : i = 1, \dots, N_d, s = 0, \dots, T_i - 1\}$.

Definition IV.8 (Persistency of Excitation) The data collection $\{\Phi_s^{(i)}\}_{(i,s) \in \mathcal{I}}$ is *persistently exciting* of order *N* if

$$\mathcal{B}_N \subseteq \mathcal{R}(\mathcal{F}_N), \quad (55)$$

where the range of $\mathcal{F}_N : \mathcal{T} \rightarrow \ell^2$ is

$$\mathcal{R}(\mathcal{F}_N) := \left\{ \sum_{(i,s) \in \mathcal{I}} g_s^{(i)} \Phi_s^{(i)} : g \in \mathcal{T} \right\},$$

i.e., the set of all ℓ^2 -convergent linear combinations of data sequences with $g \in \mathcal{T}$.

The key question for deriving the nonlinear fundamental lemma is whether any admissible *N-step trajectory* $\Phi \in \mathcal{B}_N$ can be represented as $\mathcal{F}_N g$ for some $g \in \mathcal{T}$. The PE condition (55) alone does not guarantee well-posedness in infinite dimensions, since $\mathcal{R}(\mathcal{F}_N)$ may not be closed, i.e., a sequence $\Phi^{(n)} = \mathcal{F}_N g^{(n)}$ may converge in ℓ^2 to a limit outside $\mathcal{R}(\mathcal{F}_N)$. Hence, we strengthen the PE condition as follows.

Assumption IV.9 (Quantitative PE) The data sequences $\{\Phi_s^{(i)}\}_{(i,s) \in \mathcal{I}}$ form a *frame* [25, Definition 5.1.1] for $\mathcal{B}_N$, i.e. there exist $0 < \alpha_F \leq \beta_F < \infty$ such that

$$\alpha_F \|\Phi\|_{\ell^2}^2 \leq \sum_{(i,s) \in \mathcal{I}} |\langle \Phi_s^{(i)}, \Phi \rangle_{\ell^2}|^2 \leq \beta_F \|\Phi\|_{\ell^2}^2, \quad \forall \Phi \in \mathcal{B}_N. \quad (56)$$

Under Assumption IV.9, the Hankel operator $\mathcal{F}_N$ in Definition IV.6 serves as the synthesis operator of the frame $\{\Phi_s^{(i)}\}_{(i,s) \in \mathcal{I}}$. The upper bound in (56) implies that $\{\Phi_s^{(i)}\}_{(i,s) \in \mathcal{I}}$ is a Bessel sequence with bound β_F , so $\mathcal{F}_N$ is bounded with $\|\mathcal{F}_N\|^2 \leq \beta_F$ [25, Theorem 3.2.3]. The lower bound guarantees that $\mathcal{R}(\mathcal{F}_N)$ is closed and satisfies $\mathcal{R}(\mathcal{F}_N) = \overline{\text{span}}_{\ell^2} \{\Phi_s^{(i)}\}_{(i,s) \in \mathcal{I}}$ [25, Lemma 5.2.1]. The adjoint $\mathcal{F}_N^* : \ell^2 \rightarrow \mathcal{T}$ is the analysis operator

$$[\mathcal{F}_N^* \Phi]_s^{(i)} = \langle \Phi_s^{(i)}, \Phi \rangle_{\ell^2}, \quad (i, s) \in \mathcal{I}.$$

This yields the frame operator $S_F := \mathcal{F}_N \mathcal{F}_N^* : \ell^2 \rightarrow \ell^2$ [25, eq. (5.5)], which is bounded, self-adjoint, and positive by [25, Lemma 5.1.5]. The lower frame bound gives $\langle S_F \Phi, \Phi \rangle_{\ell^2} \geq \alpha_F \|\Phi\|_{\ell^2}^2$ for all $\Phi \in \mathcal{B}_N$, so S_F is boundedly invertible on $\mathcal{B}_N$ with $\|S_F^{-1}\| \leq \alpha_F^{-1}$.

Proposition IV.10 *Assumption IV.9 implies the PE condition (55). Moreover, for every $\Phi \in \mathcal{B}_N$, the equation $\mathcal{F}_N g = \Phi$ has the explicit minimum-norm solution*

$$g = \mathcal{F}_N^* S_F^{-1} \Phi \in \mathcal{T}.$$

Proof: For any $\Phi \in \mathcal{B}_N$, S_F is boundedly invertible on $\mathcal{B}_N$ with $\|S_F^{-1}\| \leq \alpha_F^{-1}$, so $S_F^{-1} \Phi \in \ell^2$ is well defined and $g := \mathcal{F}_N^* S_F^{-1} \Phi \in \mathcal{T}$. Then

$$\mathcal{F}_N g = \mathcal{F}_N \mathcal{F}_N^* S_F^{-1} \Phi = S_F S_F^{-1} \Phi = \Phi,$$

so $\Phi \in \mathcal{R}(\mathcal{F}_N)$ and the PE condition (55) holds. ■

Remark IV.11 (Relation to Willems' fundamental lemma)

For a linear system $x_{t+1} = Ax_t + Bu_t$, $y_t = Cx_t + Du_t$, the *N-step data sequences* are defined as $\Phi_s := \text{col}(u_{[s, s+N-1]}, y_{[s+1, s+N]}) \in \mathbb{R}^{(m+p)N}$, and the truncated behavior reduces to the finite-dimensional subspace

$$\begin{aligned} \mathcal{B}_N^{\text{IO}} := \{ \Phi_s \in \mathbb{R}^{(m+p)N} \mid \exists x_0 \in \mathbb{R}^n \text{ s.t.} \\ x_{t+1} = Ax_t + Bu_t, y_t = Cx_t + Du_t \} \end{aligned}$$

of dimension $mN + n$. With a single dataset ($N_d = 1$), the Hankel operator reduces to the finite matrix $H_{N,T} \in \mathbb{R}^{(m+p)N \times T}$ with columns Φ_s , $s = 0, \dots, T-1$, and the PE condition $\mathcal{B}_N^{\text{IO}} \subseteq \mathcal{R}(H_{N,T})$ reduces to the rank condition

$$\text{rank}(H_{N,T}) = \dim(\mathcal{B}_N^{\text{IO}}) = mN + n. \quad (57)$$

This rank condition is achieved when the input is PE of order $N+n$ and (A, B) is controllable [4]. In this finite-dimensional setting, the columns of $H_{N,T}$ form a frame for $\mathcal{B}_N^{\text{IO}}$ if and only if the rank condition (57) holds, with frame bounds

$$\alpha_F = \sigma_{\min}^2(H_{N,T}), \quad \beta_F = \sigma_{\max}^2(H_{N,T}), \quad (58)$$

corresponding to the eigenvalues of the finite-dimensional frame operator $S_F = H_{N,T} H_{N,T}^\top$. The quantitative PE bound α_F coincides with the PE notion introduced in [37]. Thus Definition IV.8 generalises (57) to infinite dimensions, and Assumption IV.9 is the corresponding generalisation of the quantitative PE of [37], both fitted to the bilinear Koopman system (51).

Theorem IV.12 (Nonlinear Fundamental Lemma) *Let*

$\{\psi_{i,x}\}_{i=1}^\infty, \{\psi_{i,u}\}_{i=1}^\infty, \{\psi_{i,y}\}_{i=1}^\infty$ be Riesz bases for $\mathcal{H}_x, \mathcal{H}_u, \mathcal{H}_y$ with $\psi_{i,x} \circ F \in \mathcal{H}, \psi_{i,y} \circ h \in \mathcal{H}_x$ for all $i \in \mathbb{N}$, let Ψ_x, Ψ_y be injective, and let Assumption IV.9 hold for $\{\Phi_s^{(i)}\}_{(i,s) \in \mathcal{I}}$. Define the augmented Hankel operator $\mathcal{F}_N^w : \mathcal{T} \rightarrow \ell^2$ by $\mathcal{F}_N^w g := \sum_{(i,s) \in \mathcal{I}} g_s^{(i)} \Phi_s^{w,(i)}$, where $\Phi_s^{w,(i)} := \text{col}(\Phi_s^{(i)}, w_{[s+1, s+N]}^{(i)}) \in \ell^2$ appends the lifted output sequence to $\Phi_s^{(i)}$.

Then for any $t \in \mathbb{N}$, $\{x_{[t,t+N]}, u_{[t,t+N-1]}, y_{[t+1,t+N]}\}$ is a solution of (41) if and only if there exists $g \in \mathcal{T}$ such that

$$\mathcal{F}_N^w g = \begin{pmatrix} (z \otimes v)_{[t,t+N-1]} \\ w_{[t+1,t+N]} \end{pmatrix}. \quad (59)$$

Proof: Recall that $\mathcal{K} : \ell^2 \rightarrow \ell^2$ and $\mathcal{C} : \ell^2 \rightarrow \ell^2$ denote the coordinate representations of the composition operators C_F and C_h in corresponding Riesz bases. Define

$$\mathcal{M} : \ell^2 \rightarrow \ell^2, \quad \mathcal{M} := \begin{pmatrix} I_{\ell^2} & 0 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & I_{\ell^2} \\ \mathcal{CK} & 0 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \mathcal{CK} \end{pmatrix}, \quad (60)$$

with N identity blocks and N blocks of $\mathcal{CK}$, where $\mathcal{CK} : \ell^2 \rightarrow \ell^2$ maps the coordinate sequence of $(z_t \otimes v_t)$ to the coordinate sequence of w_{t+1} via the Riesz basis representations. Then $\Phi_s^{w,(i)} = \mathcal{M}\Phi_s^{(i)}$ and $\mathcal{F}_N^w = \mathcal{M}\mathcal{F}_N$.

($\Rightarrow$) If $\{x_{[t,t+N]}, u_{[t,t+N-1]}, y_{[t+1,t+N]}\}$ is a solution of (41), then by Theorem III.1 the lifted trajectory satisfies $z_{i+1} = \mathcal{K}(z_i \otimes v_i)$, $w_i = \mathcal{C}z_i$, $i = t, \dots, t+N-1$, so $(z \otimes v)_{[t,t+N-1]} \in \mathcal{B}_N$. By Assumption IV.9 and Proposition IV.10 there exists $g \in \mathcal{T}$ with $\mathcal{F}_N g = (z \otimes v)_{[t,t+N-1]}$. Applying $\mathcal{M}$ to both sides and using $w_{i+1} = \mathcal{CK}(z_i \otimes v_i)$ yields

$$\begin{aligned} \mathcal{F}_N^w g &= \mathcal{M}\mathcal{F}_N g = \mathcal{M}(z \otimes v)_{[t,t+N-1]} \\ &= \text{col}((z \otimes v)_{[t,t+N-1]}, w_{[t+1,t+N]}). \end{aligned}$$

($\Leftarrow$) Suppose (59) holds for some $g \in \mathcal{T}$. Splitting into top and bottom blocks gives

$$\begin{aligned} \mathcal{F}_N g &= (z \otimes v)_{[t,t+N-1]}, \\ \text{diag}(\mathcal{CK}, \dots, \mathcal{CK}) \mathcal{F}_N g &= w_{[t+1,t+N]}. \end{aligned}$$

Since $\mathcal{K} : \ell^2 \rightarrow \ell^2$ is bounded and linear, the operator $L : \ell^2 \rightarrow \ell^2$, $L\Phi_t := \text{col}(z_{i+1} - \mathcal{K}(z_i \otimes v_i))_{i=t}^{t+N-1}$ is bounded and linear as the difference of two bounded linear maps, and $\mathcal{B}_N = \ker(L)$ is therefore a closed linear subspace of ℓ^2 , see, e.g., [24, proof of Thm. 4.12]. Since each data sequence $\Phi_s^{(i)} \in \mathcal{B}_N$ by construction, and $\mathcal{B}_N$ is closed, the top block $(z \otimes v)_{[t,t+N-1]} = \mathcal{F}_N g \in \mathcal{B}_N$, so the Koopman dynamics $z_{i+1} = \mathcal{K}(z_i \otimes v_i)$ hold for $i = t, \dots, t+N-1$. The bottom block then gives $w_{i+1} = \mathcal{CK}(z_i \otimes v_i) = \mathcal{C}z_{i+1}$ for each $i = t, \dots, t+N-1$. Hence the full sequence $\{z_{[t,t+N]}, v_{[t,t+N-1]}, w_{[t+1,t+N]}\}$ is a solution of system (51), and by Theorem IV.5, injectivity of Ψ_x and Ψ_y uniquely recovers $\{x_{[t,t+N]}, u_{[t,t+N-1]}, y_{[t+1,t+N]}\}$ as a solution of (41). ■

Remark IV.13 (Data generation guidelines) For universal kernels k_x, k_u on compact X, U [31], the kernel feature maps generate function systems $\{\psi_{i,x}\}_{i=1}^\infty, \{\psi_{l,u}\}_{l=1}^\infty$ that can be chosen to form Riesz bases for $\mathcal{H}_{k_x}$ and $\mathcal{H}_{k_u}$, with tensor products $\{\psi_{i,x}\psi_{l,u}\}$ forming a Riesz basis for $\mathcal{H} = \mathcal{H}_{k_x} \otimes \mathcal{H}_{k_u}$. Using such kernels with well-distributed initial conditions $\{x_0^{(i)}\}_{i=1}^{N_d}$ in X and persistently exciting input

sequences of order N in U creates dense subspaces in ℓ^2 , which favors the construction of a frame for $\mathcal{B}_N$ and hence the satisfaction of Assumption IV.9 with $\alpha_F > 0$. Multiple data sets with independently chosen initial conditions and inputs further facilitate the required coverage of $\mathcal{B}_N$. In the finite-dimensional EDMD case, this yields a strong quantifiable PE condition for bilinear systems analogous to merging [37], the bilinear fundamental lemma [19] and its multiple data sets extension [20].

Remark IV.14 (Multi-step approximation) In the finite-dimensional case with n_z state observables and n_v input observables, the lifted trajectories $z_t \in \mathbb{R}^{n_z}$, $v_t \in \mathbb{R}^{n_v}$ and the Hankel operator reduces to the finite matrix $\mathcal{F}_N \in \mathbb{R}^{n_z n_v N \times \sum_i T_i}$. If $\mathcal{F}_N$ has full row rank, the system (59) has the minimum-norm least-squares solution

$$g^* := \mathcal{F}_N^\dagger (z \otimes v)_{[t,t+N-1]}, \quad (61)$$

where $\mathcal{F}_N^\dagger$ is the Moore-Penrose pseudoinverse, and the predicted lifted output sequence is

$$w_{[t+1,t+N]}^* = \underbrace{\left[w_{[s+1,s+N]}^{(i)} \right]_{(i,s) \in \mathcal{I}}}_{\text{finite-dim. approx. of } \text{diag}(\mathcal{CK}, \dots, \mathcal{CK})} \mathcal{F}_N^\dagger (z \otimes v)_{[t,t+N-1]}. \quad (62)$$

This generalizes EDMD to multi-step prediction directly from data, without explicitly identifying $\mathcal{K}$, and is well-suited for Koopman MPC with prediction horizon N [9].

V. ILLUSTRATIVE EXAMPLES

All results in this section were generated using a 13th Gen Intel® Core™ i7-1370P CPU running at 1.90 GHz on a Lenovo Windows Laptop and using Matlab 2023a.

A. Van der Pol Oscillator

We illustrate the developed generalized Koopman operator method for the Van der Pol oscillator. The training data set is generated by simulating the forced Van der Pol oscillator

$$\dot{x}_1 = x_2, \quad \dot{x}_2 = \mu(1 - x_1^2)x_2 - x_1 + u, \quad (63)$$

with $\mu = 1.2$, discretized at sampling time $T_s = 0.1$ s via `ode45`.

A total of $N_d = 100$ initial conditions are drawn from $X = [-2, 2]^2$ using a scrambled Halton sequence, and $N_{\text{seq}} = 10$ persistently exciting multisine input sequences of length $T_{\text{seq}} = 16$ steps, with amplitudes in $U = [-1, 1]$, are applied to each initial condition, yielding a database of $N_d \times N_{\text{seq}} \times T_{\text{seq}} = 16000$ one-step transition pairs (x_t, u_t, x_{t+1}) . From this database, $N_{\text{train}} = 15000$ training pairs and $N_{\text{val}} = 100$ validation pairs are selected using D^2 landmark sampling [38] on the joint state-input space $X \times U$. Two kernel-based EDMD variants are implemented and compared.

a) *Standard joint-kernel EDMD - kernelized KIC [10]*: The observable map lifts the joint state-input pair as

$$\Psi(x, u) = [k_\sigma([x; u], c_1), \dots, k_\sigma([x; u], c_{N_b})]^\top \in \mathbb{R}^{N_b}, \quad (64)$$

where $c_i = [x_i; u_i] \in \mathbb{R}^{n+m}$ are $N_b = 5000$ or $N_b = 10000$ randomly selected joint landmark pairs and k_σ is the inverse quadratic (IQ) kernel, i.e.,

$$k_\sigma(z, c_i) = \frac{1}{1 + \|z - c_i\|^2 / \sigma^2}, \quad z = \begin{bmatrix} x \\ u \end{bmatrix} \in \mathbb{R}^{n+m},$$

with a single bandwidth $\sigma = 1$. The regression target is $\Psi(x^+, u)$, i.e., the same joint lifting evaluated at the next state while keeping the current input, so that the identified operator $\hat{\mathcal{K}} \in \mathbb{R}^{N_b \times N_b}$ satisfies

$$\hat{\mathcal{K}} = \Psi_Z + \Psi_Z^\top (\Psi_Z \Psi_Z^\top + \gamma I)^{-1},$$

where $\Psi_Z = [\Psi(z_1), \dots, \Psi(z_N)]$ with $z_k = [x_k; u_k]$ and $\Psi_{Z^+} = [\Psi(z_1^+), \dots, \Psi(z_N^+)]$ with $z_k^+ = [x_k^+; u_k]$; $\gamma = 1e-6$.

b) *Khatri-Rao EDMD - Generalized Koopman*: The observable map instead factorizes the state and input liftings independently via

$$\Psi(x, u) = \Psi_x(x) \otimes \Psi_u(u) \in \mathbb{R}^{M_x M_u},$$

using $M_x = 100$ or $M_x = 200$ state landmarks with bandwidth $\sigma_x = 1$ and $M_u = 50$ input landmarks with bandwidth $\sigma_u = 0.54$, selected independently from the training data. The state and input kernels are defined independently on their respective spaces, with centers $c_i^x \in \mathbb{R}^n$ and $c_j^u \in \mathbb{R}^m$:

$$k_{\sigma_x}(x, c_i^x) = \frac{1}{1 + \|x - c_i^x\|^2 / \sigma_x^2},$$

$$k_{\sigma_u}(u, c_j^u) = \frac{1}{1 + \|u - c_j^u\|^2 / \sigma_u^2},$$

with $\sigma_x = 1$ and $\sigma_u = 0.54$, giving the vectors of observables

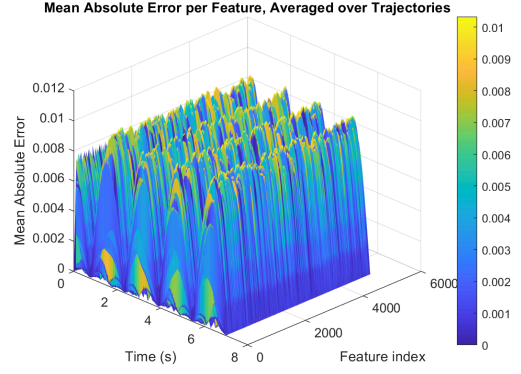
$$\Psi_x(x) = [k_{\sigma_x}(x, c_1^x), \dots, k_{\sigma_x}(x, c_{M_x}^x)]^\top \in \mathbb{R}^{M_x},$$

$$\Psi_u(u) = [k_{\sigma_u}(u, c_1^u), \dots, k_{\sigma_u}(u, c_{M_u}^u)]^\top \in \mathbb{R}^{M_u}.$$

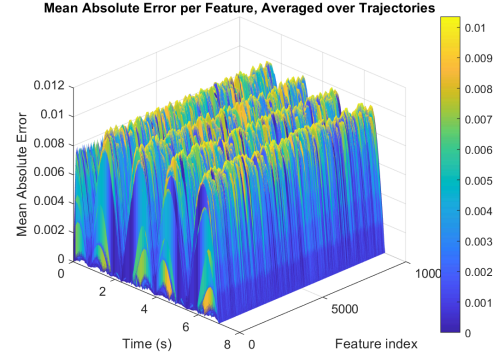
This implies that the total number of features is the same for both methods, i.e., 5000 and 10000, respectively, selected out of 15000 data points, which ensures a well-posed regression problem. The regression target is the state-only lifting $\Psi_x(x^+) \in \mathbb{R}^{M_x}$, yielding $\hat{\mathcal{K}} \in \mathbb{R}^{M_x M_u \times M_x}$, computed as in equation (39) with regularization weight $\gamma = 1e-6$.

For each method we compute the absolute error between the lifted state in the corresponding Hilbert space obtained by lifting the true trajectory of the Van der Pol oscillator versus the trajectory propagated in time by the corresponding Koopman operator, i.e., KIC and generalized Koopman, respectively. Figure 4 shows the absolute error plots for the KIC method for 5000 observables and 10000 observables, respectively. We observe that the absolute feature error does not improve noticeably despite doubling the dimension of the feature space.

Figure 5 shows the absolute error plots for the developed generalized Koopman operator for 100 and 200 state observables, respectively, and 50 input observables in all cases. The



(a) 5000 $\text{col}(x, u)$ -observables



(b) 10000 $\text{col}(x, u)$ -observables

Fig. 4: Kernelized KIC method [10]: Absolute feature error 3D plot averaged over 100 test trajectories.

dimension of the product Hilbert space is thus 5000 and 10000, respectively. As predicted by the theory developed in this paper, we observe that the absolute error consistently improves with the dimension of the feature space, which is also confirmed by the statistical plots shown in Figure 6.

B. Soft-robotic manipulator

As a second illustrative example we consider a non-input-affine nonlinear model motivated by pneumatically driven soft-robotic manipulators [39]. The state vector is $x = [\theta, \omega, p]^\top$, where θ is the bending angle, $\omega = \dot{\theta}$ is its rate, and p is the internal chamber pressure. The dynamics are

$$\begin{aligned} \dot{\theta} &= \omega, \\ \dot{\omega} &= -d\omega - k\theta + \alpha p^2, \\ \dot{p} &= a \tanh(6u) \sqrt{\max(p_s - p, 0)} - b \sqrt{\max(p, 0)} - cp, \\ y &= \ell \sin \theta, \end{aligned} \quad (65)$$

with plant parameters $d = 0.8$, $k = 2$, $\alpha = 0.35$, $a = 3$, $b = 1.1$, $c = 0.25$ and supply pressure $p_s = 5$.

The output of interest is the tip displacement with link length $\ell = 1$. Equations are integrated with the forward Euler method at sample time $T_s = 0.05$ s. The model in (65) captures three dominant dynamic phenomena of soft pneumatic actuators: elastic restoring forces ($k\theta$), viscous damping ($d\omega$), pressure-dependent bending torque (αp^2), and nonlinear

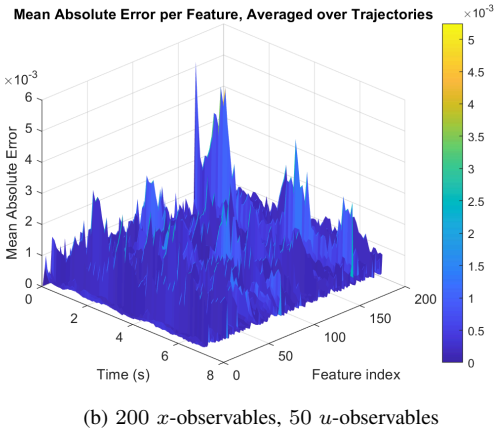
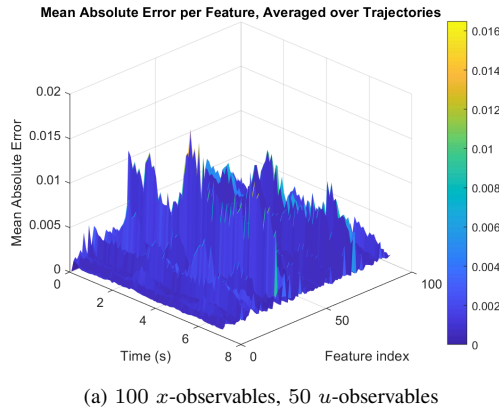


Fig. 5: Generalized Koopman: Absolute feature error 3D plot averaged over 100 test trajectories.

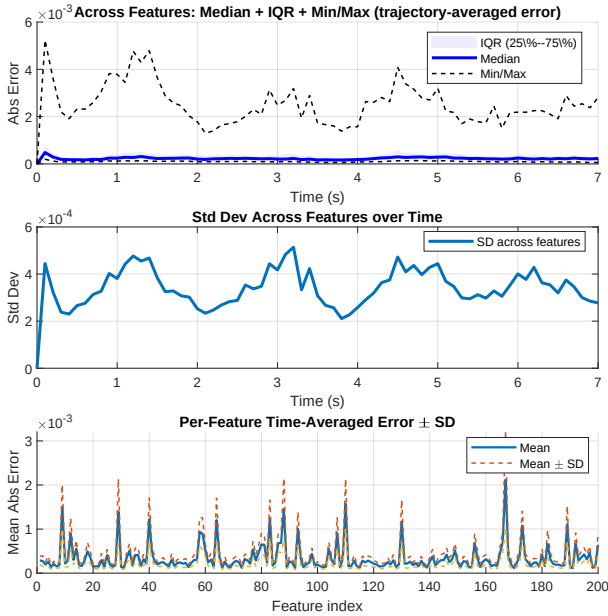


Fig. 6: Statistical analysis of the feature error for 200 x -observables, 50 u -observables.

filling/discharge through square-root flow terms. Its qualitative structure mirrors reduced-order models used in soft-robot

control applications [39].

The state, input and output spaces are defined as

$$X = \{x \in \mathbb{R}^3 \mid \theta \in [-1.5, 1.5], \omega \in [-6.0, 6.0], p \in [0, p_s]\},$$

$$U = \{u \in \mathbb{R} \mid u \in [-1, 1]\},$$

$$Y = \{y \in \mathbb{R} \mid y \in [\sin(1.5), \sin(1.5)] \approx [-0.997, 0.997]\},$$

where angles are in radians and $p_s = 5$ is the supply pressure.

We test four Koopman models that share the same $n_z = 9$ state observable vector, defined by the map $\Psi_x : X \rightarrow \mathbb{R}^{n_z}$:

$$z_t := \Psi_x(x_t) = [1, \theta, \omega, p, \sin \theta, \cos \theta, p^2, \theta \omega, \ell \sin \theta]^\top. \quad (66)$$

The dictionary (66) is constructed to capture the dominant nonlinear structure of the plant (65). The output $y = \ell \sin \theta$ is included as z_9 so that no reconstruction mapping is required.

Four choices of $\Psi_u : U \rightarrow \mathbb{R}^{n_v}$ are compared, defining $v_t := \Psi_u(u_t)$ as follows: A – *Linear*, i.e., the Koopman matrix is identified from the concatenated regressor $[z; u/\sigma_u]$; B – *Bilinear*, i.e., $\Psi_u^{(B)}(u) = [1; u]$, yielding an $(n_z \times 2)$ -block product regressor; C – *Bilinear and Odd Chebyshev*, i.e., $\Psi_u^{(C)}(u) = [1; u; T_5(u); T_7(u); T_9(u)]$, where T_n denotes the n -th Chebyshev polynomial of the first kind; D – *Bilinear and Tanh bank*, i.e., $\Psi_u^{(D)}(u) = [1; u; \tanh(4u); \tanh(8u)]$, with the input observables explicitly matching the true actuator nonlinearity, but not the actual parameter value.

For models B–D the Koopman matrix $\hat{K} \in \mathbb{R}^{n_z \times n_z n_v}$ is identified by ridge-regularised least squares on $N_{\text{train}} = 20000$ samples:

$$K = Z^+ \Phi^\top (\Phi \Phi^\top + \lambda I)^{-1},$$

where $\Phi_{:,t} = z_t \otimes v_t$ and $Z_{:,t}^+ = z_{t+1}$, consistent with Khatri-Rao EDMD (39). Input features are standardised by their training-set standard deviation prior to regression to improve numerical conditioning. Training data are generated by a mixed multi-sine / PRBS excitation (60% multi-sine, 40% PRBS, 16 sine components) from initial state $x_0 = [0.2; 0; 0.5]$.

The Koopman MPC controllers solve at each sample t :

$$\min_{U_t} \sum_{i=1}^{N_h} [Q_y(z_{t+i,9} - r_{t+i})^2 + R_u u_{t+i-1}^2 + R_{\Delta u} (\Delta u_{t+i-1})^2]$$

subject to $U_t \in U^N$, and the Koopman model state z_{t+i} is predicted using the corresponding A–D Koopman model. The cost is evaluated in the Koopman-lifted space; no reconstruction is needed because $z_9 = y$. The MPC parameters are $N_h = 15$, $Q_y = 550$, $R_u = 0.05$, $R_{\Delta u} = 1$. For the nonlinear Koopman MPC problems we use `fmincon` with SQP, warm-started from the previous solution, while for the linear Koopman MPC we use `quadprog`.

The open-loop validation input is a 125s, 16-component multi-sine signal with frequencies uniformly drawn from $[0.008, 0.64]$ Hz, random amplitudes and phases, normalised to U , which is independent of the training excitation. Closed-loop evaluation runs $N_{\text{sim}}=900$ steps (45s) on a slow piecewise-constant reference profile with smooth tanh-shaped transitions. Both open- and closed-loop simulations start at zero initial states.

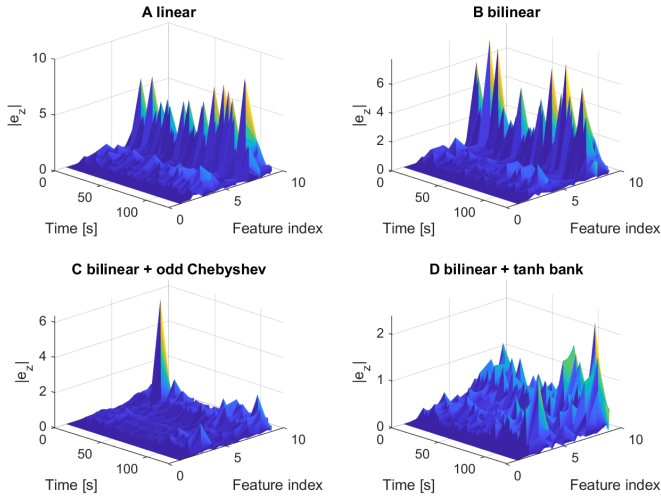


Fig. 7: 3D absolute error plot across state features for different Koopman embeddings.

Table II reports the free-run RMSE on the output y over the entire test horizon. The tanh-bank model D achieves the lowest error, consistent with its basis matching the true actuator characteristic. The odd Chebyshev model C is competitive; the bilinear model B improves over the linear-in-control baseline A, but leaves residual error.

TABLE II: Open-Loop Free-Run RMSE on Output y

Model	Input observables	RMSE
A – linear	$[u]$	0.315789
B – bilinear	$[1; u]$	0.253991
C – bil. + odd Chebyshev	$[1; u; T_5; T_7; T_9]$	0.229199
D – bil. + tanh bank	$[1; u; \tanh 4u; \tanh 8u]$	0.219148

Fig. 7 shows that the absolute feature error is decreasing with a richer observable dictionary for the input. This shows that for non-input-affine nonlinear systems, going beyond purely bilinear Koopman models can increase accuracy.

Fig. 8 shows the closed-loop output trajectories for all four Koopman MPC controllers against the reference. Table III summarises the closed-loop tracking RMSE, which is consistent with the absolute feature error results.

TABLE III: Closed-Loop Tracking RMSE and Median CPU

Model	Tracking RMSE	Median CPU [s]
A – linear	0.020783	0.000361
B – bilinear	0.016108	0.046529
C – bil. + odd Chebyshev	0.011217	0.080341
D – bil. + tanh bank	0.007468	0.099754

Note that when the reference comes closer to zero, i.e., 0.3, a smaller input is required and the input nonlinearity $\tanh$ behaves almost linearly. As a result, the linear in u Koopman MPC outperforms the bilinear in u Koopman MPC in this operating range. In contrast, the Koopman MPC based on generalized bilinear models with lifted inputs manages to consistently outperform both the linear and bilinear in u Koopman MPC over the complete tested range of operation.

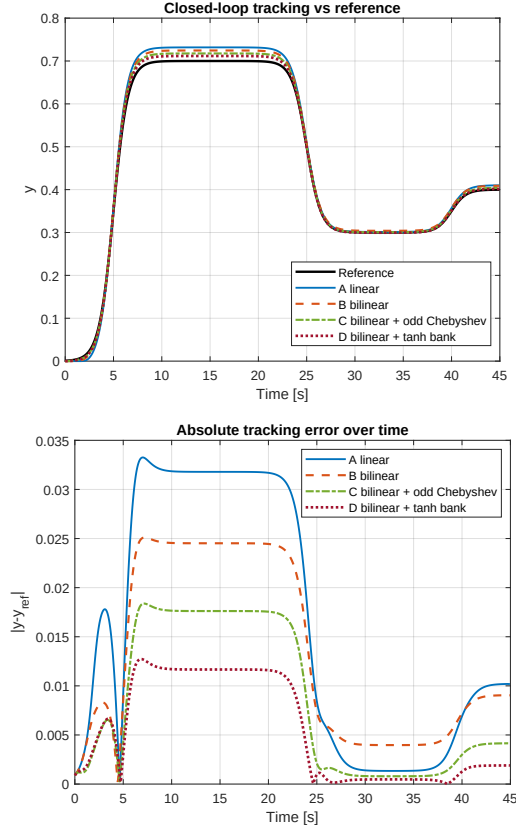


Fig. 8: A–D Koopman MPC closed-loop results. Top: reference tracking; Bottom: tracking error.

The per-step CPU times (median averaged over $N_{\text{sim}} - 1$ calls to `fmincon/quadprog`) scale roughly with the dimension of the operator. Linear and bilinear in u Koopman MPC implementations remain within real-time requirements of $T_s = 0.05$ s, while the other Koopman MPC implementations stay close to the required sampling time and could easily reach it via dedicated fast nonlinear MPC solvers (e.g., `acados`). Efficient numerical implementation is beyond the scope of this paper, however, and it will be considered in future work.

VI. CONCLUSIONS

In this paper we have presented a product Hilbert space approach to generalizing the Koopman operator for nonlinear systems with inputs. We have proven that there exists a well-defined bounded linear operator from the tensor product of two Hilbert spaces spanned by state and input observable functions, respectively, to the state Hilbert space, which yields a generalized, non-measure preserving Koopman operator for systems with inputs. Furthermore, the developed generalized Koopman operator, which leads to an exact infinite-dimensional bilinear model, was utilized to develop a nonlinear fundamental lemma. Methods for data-driven finite-dimensional approximations of the generalized Koopman operator were also indicated along with two illustrative examples that showcased the improved modeling accuracy and control performance of the developed generalized Koopman operator.

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APPENDIX I

PROOF OF THEOREM III.10

Since Assumption III.9 requires $\psi_{i,x} \circ F \in \mathcal{H}_x \otimes \mathcal{V}_u = \mathcal{H}$ for all $i \in \mathbb{N}$, the invariance condition (22) of Theorem III.6 is satisfied. Applying Theorem III.6 gives existence of $\mathcal{K} = Q_x R^{-1}$ such that $z_{t+1} = \mathcal{K}(z_t \otimes v_t)$. Substituting (35) into the definition $[Q_x]_{i,(k,l)} := \langle \psi_{i,x} \circ F, \psi_{k,x} \psi_{l,u} \rangle_{\mathcal{H}}$ and applying Fubini’s theorem yields:

$$\begin{aligned}
 [Q_x]_{i,(k,l)} &= \int_{X \times U} \psi_{i,x}(F(x, u)) \psi_{k,x}(x) \psi_{l,u}(u) d(\mu_x \otimes \mu_u) \\
 &= \langle a_i, \psi_{k,x} \rangle_{\mathcal{H}_x} \underbrace{\int_U \psi_{0,u} \psi_{l,u} d\mu_u}_{[R_u]_{0l}} \\
 &\quad + \sum_{j=1}^m \langle b_{ij}, \psi_{k,x} \rangle_{\mathcal{H}_x} \underbrace{\int_U \psi_{j,u} \psi_{l,u} d\mu_u}_{[R_u]_{jl}}, \tag{67}
 \end{aligned}$$

where $[R_u]_{0l}, [R_u]_{1l}, \dots, [R_u]_{ml}$ are the entries of the *input Gram matrix* $R_u \in \mathbb{R}^{(m+1) \times (m+1)}$. Since $\{\psi_{j,u}\}_{j=0}^m$ are linearly independent in $L^2(U, \mu_u)$ for $U \subseteq \mathbb{R}^m$ of positive

Lebesgue measure (Definition III.8), R_u is symmetric positive definite and hence invertible. The Gram matrix of the product basis $\{\psi_{k,x}\psi_{l,u}\}$ is

$$\begin{aligned} [R]_{(i,l),(k,l')} &= \langle \psi_{i,x}\psi_{l,u}, \psi_{k,x}\psi_{l',u} \rangle_{\mathcal{H}} \\ &= \langle \psi_{i,x}, \psi_{k,x} \rangle_{\mathcal{H}_x} \cdot [R_u]_{ll'} \end{aligned}$$

by Fubini's theorem, so $R = R_x \otimes R_u$ and $R^{-1} = R_x^{-1} \otimes R_u^{-1}$. Both Grams are bounded since $\|R_x^{-1}\|_{\ell^2 \rightarrow \ell^2} \leq \alpha^{-1}$ and $R_u^{-1} \in \mathbb{R}^{(m+1) \times (m+1)}$ is finite. From (67), it holds that

$$[Q_x]_{i,(k,l)} = \langle a_i, \psi_{k,x} \rangle_{\mathcal{H}_x} [R_u]_{0l} + \sum_{j=1}^m \langle b_{ij}, \psi_{k,x} \rangle_{\mathcal{H}_x} [R_u]_{jl},$$

or in matrix form:

$$Q_x = A_x \otimes (e_0^\top R_u) + \sum_{j=1}^m N_{j,x} \otimes (e_j^\top R_u),$$

where $[A_x]_{ik} := \langle a_i, \psi_{k,x} \rangle_{\mathcal{H}_x}$, $[N_{j,x}]_{ik} := \langle b_{ij}, \psi_{k,x} \rangle_{\mathcal{H}_x}$. Right-multiplying by $R^{-1} = R_x^{-1} \otimes R_u^{-1}$ and applying the mixed product property $(A \otimes B)(C \otimes D) = AC \otimes BD$ yields $\mathcal{K} = Q_x(R_x^{-1} \otimes R_u^{-1})$

$$= A_x R_x^{-1} \otimes (e_0^\top R_u R_u^{-1}) + \sum_{j=1}^m N_{j,x} R_x^{-1} \otimes (e_j^\top R_u R_u^{-1}).$$

Since $R_u R_u^{-1} = I_{m+1}$, we have $e_j^\top R_u R_u^{-1} = e_j^\top$, so the $j = 0$ block gives $\mathcal{A} := A_x R_x^{-1}$ and each $j \geq 1$ block gives $\mathcal{N}_j := N_{j,x} R_x^{-1}$, yielding the factorization $\mathcal{K} = \mathcal{A} \otimes e_0^\top + \sum_{j=1}^m \mathcal{N}_j \otimes e_j^\top$. Since $v_t = \text{col}(\psi_{0,u}(u_t), \dots, \psi_{m,u}(u_t)) = \text{col}(1, u_{1,t}, \dots, u_{m,t})$ (Definition III.8), we have $z_t \otimes v_t = \text{col}(z_t, u_{1,t}z_t, \dots, u_{m,t}z_t)$, and hence

$$z_{t+1} = \mathcal{K}(z_t \otimes v_t) = \mathcal{A}z_t + \sum_{j=1}^m \mathcal{N}_j z_t u_{j,t}.$$

Boundedness of $\mathcal{A} = A_x R_x^{-1}$ and $\mathcal{N}_j = N_{j,x} R_x^{-1}$ follows from $\|R_x^{-1}\|_{\ell^2 \rightarrow \ell^2} \leq \alpha^{-1}$ and boundedness of A_x , $N_{j,x}$, where each row $(\langle a_i, \psi_{k,x} \rangle_{\mathcal{H}_x})_{k=1}^\infty \in \ell^2$ and $(\langle b_{ij}, \psi_{k,x} \rangle_{\mathcal{H}_x})_{k=1}^\infty \in \ell^2$ by Bessel's inequality (Remark III.7 with $\mathcal{H}_u = \mathcal{V}_u$). Boundedness of $\mathcal{K}$ then follows from the factorization $\mathcal{K} = \mathcal{A} \otimes e_0^\top + \sum_{j=1}^m \mathcal{N}_j \otimes e_j^\top$.

APPENDIX II

PROOF OF PROPOSITION IV.1

Since each $\psi_{i,y} \in \mathcal{H}_y$, $i \in \mathbb{N}$, we can expand each $\psi_{i,y}$ as $\psi_{i,y} = \sum_{j=1}^\infty \langle \psi_{i,y}, \phi_j \rangle_{\mathcal{H}_y} \phi_j$. Then we have that

$$\begin{aligned} \Psi_y \circ h &= \begin{bmatrix} \psi_{1,y} \circ h \\ \psi_{2,y} \circ h \\ \vdots \end{bmatrix} = \begin{bmatrix} \sum_{j=1}^\infty \langle \psi_{1,y}, \phi_j \rangle_{\mathcal{H}_y} \phi_j \circ h \\ \sum_{j=1}^\infty \langle \psi_{2,y}, \phi_j \rangle_{\mathcal{H}_y} \phi_j \circ h \\ \vdots \end{bmatrix} \\ &= \sum_{j=1}^\infty \begin{bmatrix} \langle \psi_{1,y}, \phi_j \rangle_{\mathcal{H}_y} \\ \langle \psi_{2,y}, \phi_j \rangle_{\mathcal{H}_y} \\ \vdots \end{bmatrix} \phi_j \circ h. \end{aligned} \quad (68)$$

Since $\phi_j \circ h \in \mathcal{H}_x$ for $j \in \mathbb{N}$ and $\{\psi_{i,x}\}_{i=1}^\infty$ is an orthonormal basis in $\mathcal{H}_x$, we can expand $\phi_j \circ h$ as

$$\phi_j \circ h = \sum_{i=1}^\infty \langle \phi_j \circ h, \psi_{i,x} \rangle_{\mathcal{H}_x} \psi_{i,x}. \quad (69)$$

Substituting (69) in (68) yields

$$\begin{aligned} \Psi_y[h(x)] &= \sum_{j=1}^\infty \begin{bmatrix} \langle \psi_{1,y}, \phi_j \rangle_{\mathcal{H}_y} \\ \langle \psi_{2,y}, \phi_j \rangle_{\mathcal{H}_y} \\ \vdots \end{bmatrix} \phi_j \circ h(x) \\ &= \sum_{j=1}^\infty \begin{bmatrix} \langle \psi_{1,y}, \phi_j \rangle_{\mathcal{H}_y} \\ \langle \psi_{2,y}, \phi_j \rangle_{\mathcal{H}_y} \\ \vdots \end{bmatrix} \sum_{i=1}^\infty \langle \phi_j \circ h, \psi_{i,x} \rangle_{\mathcal{H}_x} \psi_{i,x}(x) = \\ &= \sum_{j=1}^\infty \begin{bmatrix} \langle \psi_{1,y}, \phi_j \rangle_{\mathcal{H}_y} \\ \langle \psi_{2,y}, \phi_j \rangle_{\mathcal{H}_y} \\ \vdots \end{bmatrix} [\langle \phi_j \circ h, \psi_{1,x} \rangle_{\mathcal{H}_x}, \langle \phi_j \circ h, \psi_{2,x} \rangle_{\mathcal{H}_x}, \dots] \cdot \\ &\quad \cdot \begin{bmatrix} \psi_{1,x}(x) \\ \psi_{2,x}(x) \\ \vdots \end{bmatrix} = \mathcal{C} \Psi_x(x). \end{aligned}$$

APPENDIX III

PROOF OF THEOREM IV.3

From (45) we obtain

$$\begin{bmatrix} \phi_1 \circ h \\ \phi_2 \circ h \\ \vdots \end{bmatrix} = T_y^{-1} \begin{bmatrix} \psi_{1,y} \circ h \\ \psi_{2,y} \circ h \\ \vdots \end{bmatrix},$$

which, together with (47) implies $\phi_i \circ h \in \mathcal{H}_x$ for all $i \in \mathbb{N}$. Then, from Proposition IV.1 we have that there exists a matrix $\mathcal{C}$ such that (48) holds. Multiplying both sides in (48) with a row vector function $[\psi_{1,x}, \psi_{2,x}, \dots]$ and integrating each element of the resulting matrices of functions over X , via a similar reasoning as in the proof of Theorem III.6, we obtain

$$\begin{aligned} &\begin{bmatrix} \langle \psi_{1,y} \circ h, \psi_{1,x} \rangle_{\mathcal{H}_x} & \langle \psi_{1,y} \circ h, \psi_{2,x} \rangle_{\mathcal{H}_x} & \dots \\ \langle \psi_{2,y} \circ h, \psi_{1,x} \rangle_{\mathcal{H}_x} & \langle \psi_{2,y} \circ h, \psi_{2,x} \rangle_{\mathcal{H}_x} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix} \\ &= \mathcal{C} \begin{bmatrix} \langle \psi_{1,x}, \psi_{1,x} \rangle_{\mathcal{H}_x} & \langle \psi_{1,x}, \psi_{2,x} \rangle_{\mathcal{H}_x} & \dots \\ \langle \psi_{2,x}, \psi_{1,x} \rangle_{\mathcal{H}_x} & \langle \psi_{2,x}, \psi_{2,x} \rangle_{\mathcal{H}_x} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix}, \end{aligned}$$

and thus, $Q_y = \mathcal{C} R_x$. Since $\{\psi_{i,x}\}_{i=1}^\infty$ is a Riesz basis in $\mathcal{H}_x$ and thus, its Gram matrix R_x is invertible, yields $\mathcal{C} = Q_y R_x^{-1}$.



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